

IP_3516_HS_MEM

USB2.0 HS/FS/LS USB device and host IP

(low gate-count)

Integration Guide

Version: 0.0.1

Issue Date: 05/03/26

Issued by: NXP



Revision history:

Version:	Date:	Changes:	Author:	Reviewed by:
Draft 0.0.1	05/03/26	Initial version for Caliptra-SS Integration	NXP	

1. Introduction

1.1 Purpose

This document describes the technical specifications of IP_3516_HS_MEM_SMS, the USB 2.0 High-speed/Full-speed/Low-speed Host/ Device Controller. It explains the IP architecture, the intended use of the IP, parameters and deliverables.

The IP_3516_HS_MEM_SMS is made up of IP_3511_hs (USB 2.0 high-speed & full-speed device core), IP_3515_hs (USB high-speed, full-speed & low-speed host core) and the usb20_phy_hdo1p_c090lp (USB2.0 PHY with UTMI+ interface). The IP can switch between host and device modes under software control.

1.2 Scope

This document should serve as a reference for the integration engineers and firmware developers that make use of this IP.

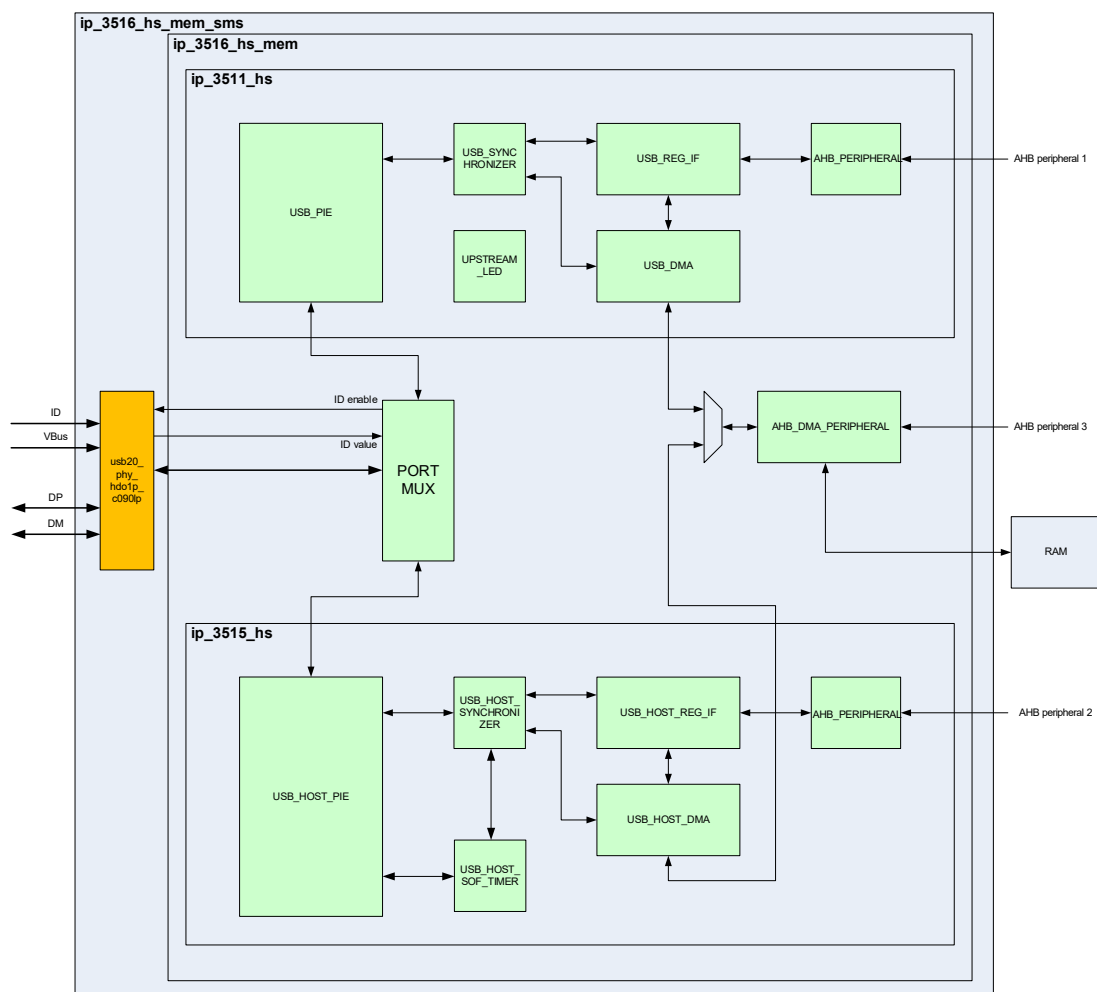
1.3 References

[AMBA_SPEC] IHI-0011A - AMBA specification Rev2.0, May 13th, 1999, <http://www.arm.com>

Table 1. : References

2. Architecture overview

The picture below gives an overview of the block diagram. The block contains 3 AHB peripheral interfaces. One interface is used to access the USB device registers (AHB peripheral 1), a second one is used to access the USB host registers (AHB peripheral 2) and a third one is used to get direct access to the RAM memory where all USB data and control information is stored (AHB peripheral 3). It is also possible to use this RAM memory as generic memory when the USB is not functional. RAM and PHY are shown only to illustrate the overall integrated block and are not part of the IP. The USB controller is designed to operate exclusively with an 8-bit UTMI+ interface. Higher-width UTMI+ configurations are not supported.



3. Hardware Interface

3.1 Pin Description

The Table below indicates all the interface signals of the USB2.0 High-Speed, Full-Speed and Low-Speed host/device IP.

Name	Dir	Width	Clock domain	Description
UTMI+ PHY signals				
utmi_clk	in	1		This is the clock input from the UTMI+ PHY. 60MHz.
utmi_rxddata	in	8	utmi_clk	8-bit parallel received data input from the UTMI+ PHY.
utmi_rxvalid	in	1	utmi_clk	This signal indicates that the utmi_rxddata bus contains the valid data.
utmi_rxactive	in	1	utmi_clk	This signal indicates that the receiver has detected the active SYNC.
utmi_rxerror	in	1	utmi_clk	This signal indicates that there is error in received data.
utmi_txdata	out	8	utmi_clk	8-bit parallel transmit data output to the UTMI+ PHY.
utmi_txvalid	out	1	utmi_clk	This signal indicates that the utmi_txdata bus is valid.
utmi_txready	in	1	utmi_clk	This signal indicates that PHY is ready to accept data.
utmi_reset	out	1	utmi_clk	This signal resets UTMI+ PHY.
utmi_suspendm	out	1	utmi_clk	This signal is used to shut down the USB functional blocks for the suspend/resume operations. While suspended, the utmi_termselect signal must be set to ensure that the 1.5kΩ pull-up resistor is connected on DP or DM. 0: Suspend mode. 1: Normal mode.
utmi_xcvrselect	out	2	utmi_clk	This signal is used to select the FS or HS transceiver. 2'b00: HS transceiver. 2'b01: FS transceiver. 2'b10: LS transceiver. 2'b11: Send a LS packet on FS bus or receive a LS packet.
utmi_termselect	out	1	utmi_clk	This signal is used to select the FS/LS or HS termination. 0: HS termination is enabled. 1: FS/LS termination is enabled.

Integration Guide

utmi_opmode	out	2	utmi_clk	This signal selects various operation modes. 00: Normal operation (The UTMI+ transceiver automatically appends the SYNC and EOP patterns). 01: Non-driving. 10: Disable the bit-stuffing and NRZI encoding. 11: Normal operation without the automatic generation of SYNC and EOP.
utmi_linestate	in	2	utmi_clk	These signals reflect the current state of DP and DM signals. These signals are synchronized to CLK when they are not suspended. Example for FS mode: DM DP Description 0 0 0: SE0 0 1 1: J 1 0 2: K 1 1 3: SE1
utmi_vcontrol	out	4	utmi_clk	Vendor-defined 4-bit parallel output bus. Default value is 4'h0. VCONTROL [3:0] indicates a specific test mode for USB use.
utmi_vcontrolloadm	out	1	utmi_clk	Assert this signal to loads the vendor control register. 0: Load the vendor-controlled register 1: Normal operation.
utmi_vstatus	in	8	utmi_clk	This signal may be tied to 0 when unused. If utilized, the connected value can be accessed through the host/device registers.
utmi_hostdisconnect	in	1	utmi_clk	If there is no peripheral connected, the signal will be 1'b1. If a peripheral is connected, then the value of this signal will be 1'b0. It is only valid when DPPULLDOWN and DMPULLDOWN are 1'b1.
utmi_id_enable	out	1	host_ahb_s_hclk	1: ID pin sampling is enabled. 0: ID pin sampling is disabled.
utmi_id_value	in	1	Async	1: The connected plug is a mini-B plug. 0: The connected plug is a mini-A plug.
utmi_dppulldown	out	1		The signal enables the pull-down resistor on the DP line. 1'b0: Pull-down resistor not connected to DP. 1'b1: Pull-down resistor connected to DP. Driven internally as 1'b0 for device and 1'b1 for Host
utmi_dmpulldown	out	1	Async fixed value	The signal enables the pull-down resistor on the DM line. 1'b0: Pull-down resistor not connected to DM.

Integration Guide

				1'b1: Pull-down resistor connected to DM. Driven internally as 1'b0 for device and 1'b1 for Host
chrgvbus	out	1	dev_ahbs_hclk	The signal enables charging Vbus. 1b: charge Vbus through a resistor (this has to be active for at least 30 ms) 0b: do not charge Vbus through a resistor
dischrgvbus	out	1	dev_ahbs_hclk	The signal enables discharging Vbus. 1b: discharge Vbus through a resistor (this must be active for at least 50 ms) 0b: do not discharge Vbus through a resistor
avalid	in	1	async	Indicates if the session for an A-peripheral is valid ($0.8V < V_{th} < 2V$). 0b: $V_{bus} < 0.8V$ 1b: $V_{bus} > 2V$
sessend	in	1	async	Indicates if the voltage on Vbus ($0.2V < V_{th} < 0.8V$). 1b: $V_{bus} < 0.2V$ 0b: $V_{bus} > 0.8V$
pdcom	out	1	utmi_clk	This signal indicates USB PHY to enter in power down mode. It can be controlled by software or hardware.
AHB peripheral interface for the device				
dev_ahbs_hclk	in	1		AHB clock for USB device
dev_ahbs_hresetn	in	1		Active low AHB bus reset for USB device. Should be synchronised with dev_ahbs_hclk
dev_ahbs_hsel	in	1	dev_ahbs_hclk	Peripheral select signal indicating that the current transfer is intended for this peripheral
dev_ahbs_hwdata	in	32	dev_ahbs_hclk	32-bit write data to peripheral
dev_ahbs_haddr	in	4	dev_ahbs_hclk	AHB peripheral address signal
dev_ahbs_hwrite	in	1	dev_ahbs_hclk	Transfer direction signal to peripheral indicating write transfer (high) or read transfer (low)
dev_ahbs_hreadyin	in	1	dev_ahbs_hclk	Transfer done signal to peripheral indicating that a transfer has finished on the bus
dev_ahbs_htrans	in	2	dev_ahbs_hclk	Transfer type signal to peripheral indicating the type of the current transfer which can be non-sequential, sequential, idle or busy.

Integration Guide

dev_ahbs_hrddata	out	32	dev_ahbs_hclk	32-bit read data from peripheral
dev_ahbs_hresp	out	2	dev_ahbs_hclk	Transfer response signal from peripheral providing additional information on the status of a transfer. The response indicated by the peripheral will be always OKAY.
dev_ahbs_hreadyout	out	1	dev_ahbs_hclk	Transfer done signal from peripheral indicating that a transfer has finished on the bus
<i>AHB peripheral interface for USB2.0 host mode</i>				
host_ahbs_hclk	in	1		AHB clock for USB host
host_ahbs_hresetn	in	1	host_ahb_s_hclk	Active low AHB bus reset for USB host. Should be synchronised with host_ahbs_hclk
host_ahbs_hsel	in	1	host_ahb_s_hclk	Peripheral select signal indicating that the current transfer is intended for this peripheral
host_ahbs_hwdata	in	32	host_ahb_s_hclk	32-bit write data to peripheral
host_ahbs_haddr	in	4	host_ahb_s_hclk	AHB peripheral address signal
host_ahbs_hwrite	in	1	host_ahb_s_hclk	Transfer direction signal to peripheral indicating write transfer (high) or read transfer (low)
host_ahbs_hreadyin	in	1	host_ahb_s_hclk	Transfer done signal to peripheral indicating that a transfer has finished on the bus
host_ahbs_htrans	in	2	host_ahb_s_hclk	Transfer type signal to peripheral indicating the type of the current transfer which can be non-sequential, sequential, idle or busy.
host_ahbs_hrddata	out	32	host_ahb_s_hclk	32-bit read data from peripheral
host_ahbs_hresp	out	2	host_ahb_s_hclk	Transfer response signal from peripheral providing additional information on the status of a transfer. The response indicated by the peripheral will be always OKAY.
host_ahbs_hreadyout	out	1	host_ahb_s_hclk	Transfer done signal from peripheral indicating that a transfer has finished on the bus
<i>AHB peripheral interface for RAM access</i>				

Integration Guide

ahbs_dma_hclk	in	1		AHB clock for RAM access interface
ahbs_dma_hresetn	in	1	ahbs_dma_hclk	Active low AHB bus reset for RAM access. Should be synchronised with ahbs_dma_hclk
ahbs_dma_hsel	in	1	ahbs_dma_hclk	Peripheral select signal indicating that the current transfer is intended for this peripheral
ahbs_dma_hwdata	in	32	ahbs_dma_hclk	32-bit write data to Peripheral
ahbs_dma_haddr	in	(RAM_ADDR_WIDT H + 5)	ahbs_dma_hclk	AHB peripheral address signal
ahbs_dma_hwrite	in	1	ahbs_dma_hclk	Transfer direction signal to peripheral indicating write transfer (high) or read transfer (low)
ahbs_dma_hreadyin	in	1	ahbs_dma_hclk	Transfer done signal to peripheral indicating that a transfer has finished on the bus
ahbs_dma_htrans	in	2	ahbs_dma_hclk	Transfer type signal to peripheral indicating the type of the current transfer which can be non-sequential, sequential, idle or busy.
ahbs_dma_hrdata	out	32	ahbs_dma_hclk	32-bit read data from peripheral
ahbs_dma_hresp	out	2	ahbs_dma_hclk	Transfer response signal from peripheral providing additional information on the status of a transfer. The response indicated by the peripheral will be always OKAY.
ahbs_dma_hreadyout	out	1	ahbs_dma_hclk	Transfer done signal from peripheral indicating that a transfer has finished on the bus
ahbs_dma_hsize	in	3	ahbs_dma_hclk	Transfer size signal to peripheral indicating the size of the transfer.
ahbs_dma_hburst	in	3	ahbs_dma_hclk	Transfer size signal from controller indicating the transfer is part of a burst
RAM interface				
mem_q	in	64	ahbs_dma_hclk	RAM read data bus

Integration Guide

mem_d	out	64	ahbs_dm a_hclk	RAM write data bus
mem_cs	out	1	ahbs_dm a_hclk	RAM enable signal
mem_a	out	C_RA M_AD DRWI DTH	ahbs_dm a_hclk	RAM address bus C_RAM_ADDRWIDTH: Defines the size of the RAM
mem_web_out	out	1	ahbs_dm a_hclk	Active low write enable signal
mem_bsel	out	64	ahbs_dm a_hclk	In case of a read or a write, these bits indicate if the corresponding data bits must be read/written mem_dword_en(0) indicates if bits (31:0) must be accessed
<i>System interface for USB2.0 host and device</i>				
dev_usb_int_req_irq	out	1	dev_ahbs _hclk	Active high interrupt line from USB2.0 device
dev_usb_int_req_fiq	out	1	dev_ahbs _hclk	Active high fast interrupt line from USB2.0 device
host_usb_int_req_irq	out	1	host_ahb s_hclk	Active high interrupt line from USB2.0 host
dev_usbframetoggle	out	1	utmi_clk	1kHz output clock from USB device based on the incoming SOF tokens
dev_usb_needclk	out	1	utmi_clk	Indicates if the USB clock must be turned on or off. 0b: USB device is in suspend mode and REFCLK clock can be turned off. 1b: USB device is operational and REFCLK clock must be on.
host_usb_needclk	out	1	utmi_clk	Indicates if the USB clock must be turned on or off. 0b: USB device is in suspend mode and REFCLK clock can be turned off. 1b: USB device is operational and REFCLK clock must be on.
dev_sys_donotwakeup_n	in	1	async	Extra external user "do not wake-up" signal for silicon test mode. Asserting this signal (active low) will result in forcing the USB device in low power mode and preventing it to wake up on any condition.
host_sys_donotwakeup_n	in	1	async	Extra external user "do not wake-up" signal for silicon test mode. Asserting this signal (active low) will result in forcing the USB host in low power mode

Integration Guide

				and preventing it to wake up on any condition.
dev_sys_wakeup_n	in	1	async	Extra external user wake-up signal for device mode; asserting this signal (active low) will result in exiting the low power mode; input to asynchronous control logic
host_usb_overcurrent_n	in	1	async	Port power fault indicating over-current condition; this signal monitors over-current on the USB bus (external circuitry required to detect over-current condition)
host_usb_portindicator	out	2	host_ahb_s_hclk	Port indicator LED control outputs for use by an external LED driving circuit. 00b = Off 01b = Amber 10b = Green 11b = Undefined
host_usb_portpower	out	1	host_ahb_s_hclk	VBus drive signal (towards external charge pump or power management unit). Indicates that Vbus must be driven (active high)
usb_token_length	out	7	utmi_clk	This bus indicates the length of the received FS SOF token (from start of packet until end of packet). This indicated in number of REFCLK clock cycles. This bus can be used to tune the REFCLK input to the required frequency when running in full-speed mode.
token_length_counter	in	7	utmi_clk	This signal can be tied to 0
dev_sys_utmi_clkin_lock	in	1	utmi_clk	Can be tied to utmi_suspendm
host_sys_utmi_clkin_lock	in	1	utmi_clk	Can be tied to utmi_suspendm
Configuration status indicator				
USB_VBus	in	1		Bus power is present.
vbuscomp_on	out	1	utmi_clk	Enable Analog Vbus comparators
ULPI signals				
ulpi_clk	in	1		ULPI interface clock running at 60MHz

Integration Guide

ulpi_rxddata	in	8	ulpi_clk	Input data bus from ULPI transceiver
ulpi_txdata	out	8	ulpi_clk	Output data bus to ULPI transceiver
ulpi_txenable	out	1	ulpi_clk	Output enable signal that can be used to generate a bidirectional data bus combining ulpi_rxddata and ulpi_txdata
ulpi_dir	in	1	ulpi_clk	ULPI Direction. Controls the direction of the data bus.
ulpi_stp	out	1	ulpi_clk	ULPI Stop. The controller asserts this signal for 1 clock cycle to stop the data stream currently on the bus.
ulpi_nxt	in	1	ulpi_clk	ULPI Next. The PHY asserts this signal to throttle the data.
ulpi_ddr_sel	in	1	ulpi_clk	This bit indicates if the interface is DDR(1b) or SDR(0b). In case of DDR, the data is clocked on both edges of the ulpi_clk and only the 4 lsb bits on the data busses are used.

3.2 Detailed timing of interfaces

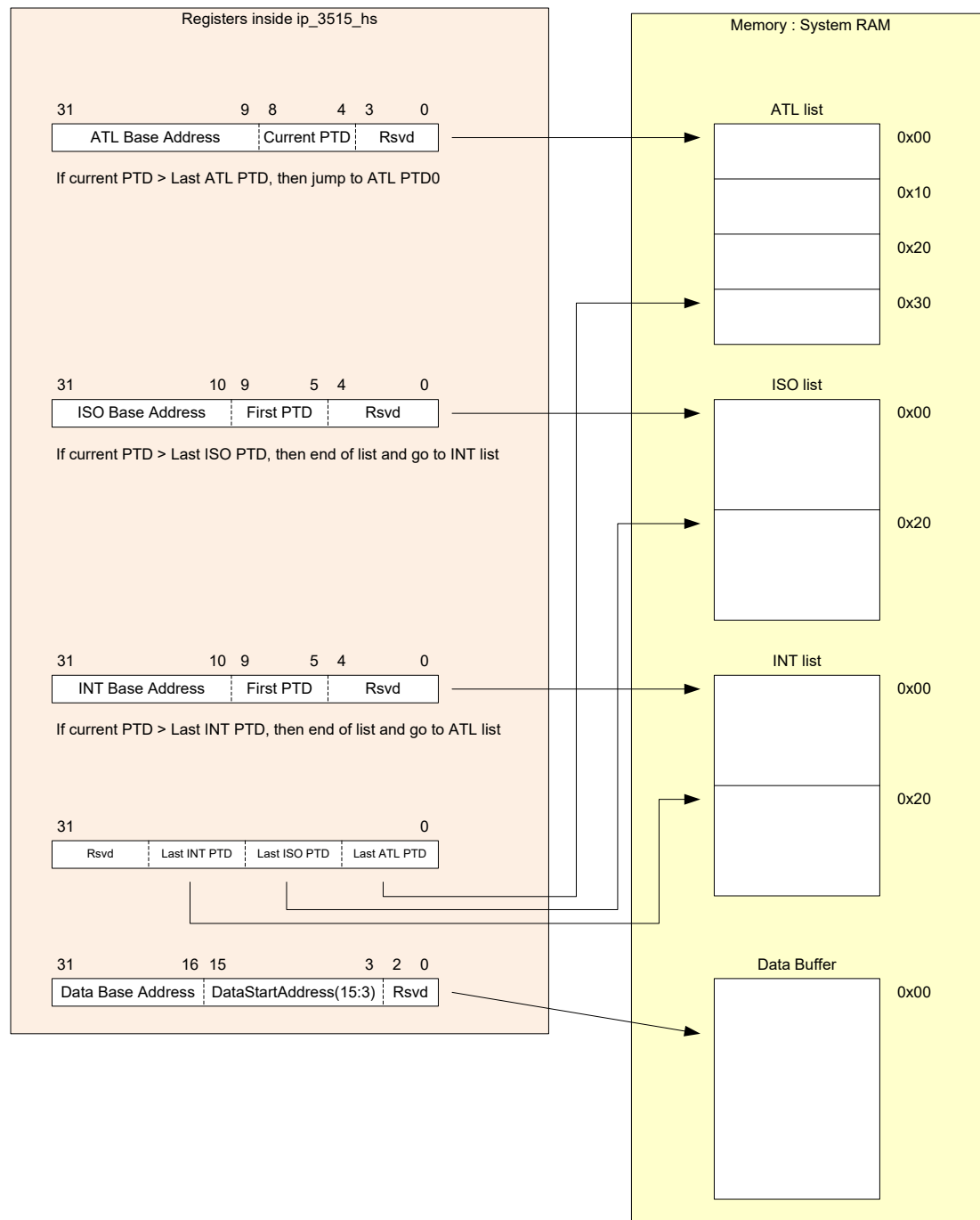
The detailed timing of the AHB interface is available in [AMBA_SPEC]

4. Software Interface

4.1 USB high-speed, full-speed and low-speed host (IP_3515_HS)

The AHB peripheral interface of the USB host must be used to access the registers as well as to configure the mode of the port (host mode or device mode).

The picture below shows which part of the data is stored in registers and what is in system memory. The registers are explained in section 4.1.1. The variables indicated in *italic* are values that can be defined at HW compile time. They are explained in section .



4.1.1 Register overview

This section gives an overview of the register set used to control the hardware functionalities of the USB Full Speed Device Controller IP. All register addresses listed below are offsets. The base address of this IP block is system-dependent and must be defined by the integrator.

Offset	Register name	Access	Description
0x00	CAPLENGTH / CHIPID	RO	This register contains the offset value towards the start of the operational register space and the version number of the IP block
0x04	HCSPARAMS	RO	Host Controller Structural Parameters
0x08	HCCPARAMS	RO	Host Controller Capability Parameters
0x0C	FLADJ/FRINDEX	RW	Frame Length Adjustment
0x10	ATL PTD Base Address	RW	Memory base address where ATL PTD0 is stored
0x14	ISO PTD Base Address	RW	Memory base address where ISO PTD0 is stored
0x18	INT PTD Base Address	RW	Memory base address where INT PTD0 is stored
0x1C	Data Payload Base Address	RW	Memory base address that indicates the start of the data payload buffers.
0x20	USBCMD	RW	USB Command register
0x24	USBSTS	RWC	USB Interrupt Status register
0x28	USBINTR	RW	USB Interrupt Enable register
0x2C	PORTSC1	RW	Port Status and Control register
0x30	ATL PTD Done Map	RWC	Done map for each ATL PTD
0x34	ATL PTD Skip Map	RW	Skip map for each ATL PTD
0x38	ISO PTD Done Map	RWC	Done map for each ISO PTD
0x3C	ISO PTD Skip Map	RW	Skip map for each ISO PTD
0x40	INT PTD Done Map	RWC	Done map for each INT PTD
0x44	INT PTD Skip Map	RW	Skip map for each INT PTD
0x48	Last PTD in use	RW	Marks the last PTD in the list for ISO, INT and ATL
0x4C	UTMI+/ULPI debug	RW	Register to read/write registers in the attached USB PHY
0x50	OTG Port Mode	RW	Controls the port if it is attached to the host block or the device block.

4.1.2 Register details

4.1.2.1 CAPLENGTH/CHIPID register (Address Offset = 0x00)

Reset value: 0x01mm0010 – *mm* is the minor version of the IP

Bit	Symbol	Access	Description
7:0	CAPLENGTH	RO	Capability Length: This is used as an offset. It is added to the register base to find the beginning of the operational register space
15:8	R	RO	Reserved
31:16	CHIPID	RO	Chip identification indicates major and minor revision of the IP - [31:24] = Major revision - [23:16] = Minor revision Major revisions used: 0x01: USB2.0 Hi-speed host

4.1.2.2 HCSPARAMS register (Address Offset = 0x04)

Reset value: 0x000g00g1 – *g* is defined by the setting of the generic map of the IP

Bit	Symbol	Access	Description
3:0	N_PORTS	RO	This register specifies the number of physical downstream ports implemented on this host controller. This is fixed to 0x1 for this IP.
4	PPC	RO	This field indicates whether the host controller implementation includes port power control. The value of this bit is controlled by the generic <i>C_PORTPOWER_CONTROL</i>
15:5	R	RO	Reserved
16	P_INDICATOR	RO	This bit indicates whether the ports support port indicator control. The value of this bit is controlled by the generic <i>C_PORT_INDICATORS</i> .
19:17	R	RO	Reserved
23:20	DPN	RO	This field identifies which of the host controller ports is the debug port. In the current version of the IP, this is set to 0x0 as there is no debug port implemented.
31:24	R	RO	Reserved

4.1.2.3 HCCPARAMS register (Address Offset = 0x08)

Reset value: 0x00020006

Bit	Symbol	Access	Description
0	R	RO	Reserved
1	PFLF	RO	Programmable Frame List Flag If this bit is cleared, the system software must use a frame list length of 1024 elements with this host controller. If PFLF is set, the system software can specify and use a smaller frame list and configure the host through the Frame List Size (FLS) field of the USBCMD register This bit is fixed to 1b for this IP
2	ASPC	RO	Asynchronously Schedule Park Capability. If this bit is set to logic1, the host controller supports the park feature for high-speed Transfer Descriptors in the Asynchronous Schedule. This bit is fixed to 1b for this IP
16:3	R	RO	Reserved
17	LPMC	RO	Link Power Management Capability. This indicates host controller support for the Link Power Management L1 state and associated PORTSC Suspend using L1, Suspend status and Device Address fields.
31:18	R	RO	Reserved

4.1.2.4 FLADJ register (Address Offset = 0x0C)

Reset value : 0x00000020

Bit	Symbol	Access	Description
5:0	FLADJ	RW	Frame Length Timing Value. Each decimal value change to this register corresponds to 16 high-speed bit times. The SOF cycle time (number of SOF counter clock periods to generate a SOF micro-frame length) is equal to 59488 + value in this field. The default value is decimal 32 (20h), which gives a SOF cycle time of 60000.
15:6	R	RO	Reserved
29:16	FRINDEX	RW	Frame Index : Bits 29 down to 19 in this register are used for the frame number field in the SOF packet. The value in this field is incremented by one every 125us (independent of the speed of the attached device) Software is only allowed to update this field when the Run/Stop bit is set to zero.
31:30	R	RO	Reserved

4.1.2.5 ATL PTD BaseAddress (Address Offset = 0x10)

Reset value : 0x00000000

Bit	Symbol	Access	Description
3:0	R	RO	Reserved
8:4	ATL_CUR	RW	This indicates the current PTD that is used by the hardware when it is processing the ATL list
31:9	ATL_BASE	RW	Base address to be used by the hardware to find the start of the ATL list.

The hardware will only use the least significant bits of this register to find the correct location in RAM. E.g. if the RAM is 4kBytes big, bits 11 downto 0 of this register will be used to find the correct byte address.

4.1.2.6 ISO PTD BaseAddress (Address Offset = 0x14)

Reset value : 0x00000000

Bit	Symbol	Access	Description
4:0	R	RO	Reserved
9:5	ISO_FIRST	RW	This indicates the first PTD that is used by the hardware when it is processing the ISO list
31:10	ISO_BASE	RW	Base address to be used by the hardware to find the start of the ISO list.

The hardware will only use the least significant bits of this register to find the correct location in RAM. E.g. if the RAM is 4kBytes big, bits 11 downto 0 of this register will be used to find the correct byte address.

4.1.2.7 INT PTD BaseAddress (Address Offset = 0x18)

Reset value : 0x00000000

Bit	Symbol	Access	Description
4:0	R	RO	Reserved
9:5	INT_FIRST	RW	This indicates the first PTD that is used by the hardware when it is processing the INT list
31:10	INT_BASE	RW	Base address to be used by the hardware to find the start of the INT list.

The hardware will only use the least significant bits of this register to find the correct location in RAM. E.g. if the RAM is 4kBytes big, bits 11 downto 0 of this register will be used to find the correct byte address.

4.1.2.8 Data Payload BaseAddress (Address Offset = 0x1C)

Reset value : 0x00000000

Bit	Symbol	Access	Description
15:0	R	RO	Reserved

Integration Guide

31:16	DAT_BASE	RW	Base address to be used by the hardware to find the start of the data payload section. The hardware will only use the least significant bits required for addressing the correct location in RAM.
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4.1.2.9 USBCMD register (Address Offset = 0x20)

Reset value : 0x00000000

Bit	Symbol	Access	Description
0	RS	RW	Run/Stop : 1b = Run. The host controller executes the schedule 0b = Stop. If this bit is set to 1b, the USB clock will be always on.
1	HCRESET	RW	Host Controller Reset : This control bit is used by the software to reset the host controller
3:2	FLS	RW	Frame List Size : This field specifies the size of the frame list. This field is used to control when the frame list roll over interrupt bit must be set. 00b 1024 elements 01b 512 elements 10b 256 elements 11b Reserved
6:4	R	RO	Reserved
7	LHCR	RW	Light Host Controller Reset : This bit allows the driver software to reset the host controller without affecting the state of the ports.
8	ATL_EN	RW	ATL List enabled. When this bit is set, hardware will process the ATL list.
9	ISO_EN	RW	ISO List enabled. When this bit is set, hardware will process the ISO list.
10	INT_EN	RW	INT List enabled. When this bit is set, hardware will process the INT list.
23:11	R	RO	Reserved
27:24	HIRD	RW	Host-Initiated Resume Duration. This field is used by system software to specify the minimum amount of time the host controller will drive the K-state during a host-initiated resume from a LPM state (e.g. L1), and is conveyed to each LPM-enabled device (via the HIRD bits within an LPM token's bmAttributes field) upon entry into a low-power state. The host controller is required to drive resume signaling for at least the amount of time specified in the HIRD value conveyed to the device during any proceeding host-initiated resume. A host controller is not required to observe this requirement during device-initiated resumes. Encoding for this field is identical to the definition for the similarly named HIRD field within an LPM token, specifically: a value 0000b equals 50us and each

Integration Guide

			additional increment adds 75us. E.g. 0001b equals 125us and a value 1111b equals 1175us.
28	LPM_RWU	RW	bRemoteWake field. This bit indicates if the device is enabled for doing a Remote Wakeup when it is in the L1 suspend state.
31:29	R	RO	Reserved

4.1.2.10 USBSTS register (Address Offset = 0x24)

Reset value : 0x00000000

Bit	Symbol	Access	Description
1:0	R	RO	Reserved
2	PCD	RWC	Port Change Detect : The host controller sets this bit to logic 1 when - any port has a change bit transition from a zero to a one or - a Force Port Resume bit transition from a zero to a one as a result of a J-K transition detected on a suspended port or - the ID pin value changes or - an LPM token has been transmitted to enter LPM L1 suspend state. Software must write a one to clear the bit
3	FLR	RWC	Frame List Rollover : The host controller sets this bit to logic 1 when the frame list index rolls over its maximum value to zero Software must write a one to clear the bit
15:4	R	RO	Reserved
16	ATL_IRQ	RWC	ATL IRQ: Indicates that an ATL PTD (with I-bit set) was completed. The hardware interrupt line will be asserted if the respective enable bit in the USBINTR register is set. 0 — No ATL PTD event occurred 1 — ATL PTD event occurred Software must write a one to clear the bit
17	ISO_IRQ	RWC	ISO IRQ: Indicates that an ISO PTD (with I-bit set) was completed. The hardware interrupt line will be asserted if the respective enable bit in the USBINTR register is set. 0 — No ISO PTD event occurred 1 — ISO PTD event occurred Software must write a one to clear the bit
18	INT_IRQ	RWC	INT IRQ: Indicates that an INT PTD (with I-bit set) was completed, The hardware interrupt line will be asserted if the respective enable bit in the USBINTR register is set. 0 — No INT PTD event occurred 1 — INT PTD event occurred Software must write a one to clear the bit

Integration Guide

19	SOF_IRQ	RWC	SOF interrupt : Every time when the host sends a Start of Frame token on the USB bus, this bit is set. Software must write a one to clear the bit.
31:20	R	RO	Reserved

4.1.2.11 USBINTR register (Address Offset = 0x28)

Reset value : 0x00000000

Bit	Symbol	Access	Description
1:0	R	RO	Reserved
2	PCDE	RW	Port Change Detect Interrupt Enable : If this enable bit is set to one and the corresponding USBSTS bit is set to one, a hardware interrupt is generated.
3	FLRE	RW	Frame List Rollover Interrupt Enable: If this enable bit is set to one and the corresponding USBSTS bit is set to one, a hardware interrupt is generated.
15:4	R	RO	Reserved
16	ATL_IRQ_E	RW	ATL IRQ Enable bit
17	ISO_IRQ_E	RW	ISO IRQ Enable bit
18	INT_IRQ_E	RW	INT IRQ Enable bit
19	SOF_E	RW	SOF Interrupt Enable bit
31:20	R	RO	Reserved

4.1.2.12 PORTSC1 register (Address Offset = 0x2C)

Reset value : 0x00000000

Bit	Symbol	Access	Description
0	CCS	RO	Current Connect Status : Logic 1 indicates a device is present on the port. Logic 0 indicates no device is present This field is zero if Port Power is zero
1	CSC	RWC	Connect Status Change : Logic 1 means that the value of CCS has changed. Logic 0 means no change. This field is zero if Port Power is zero SW must write a logic 1 to clear the bit
2	PED	RW	Port Enabled/Disabled. Logic 1 means port enabled. Logic 0 means disabled. This field is zero if Port Power is zero
3	PEDC	RWC	Port Enabled/Disabled Change : Logic 1 means that the value of PED has changed. Logic 0 means no change. This field is zero if Port Power is zero SW must write a logic 1 to clear the bit

Integration Guide

4	OCA	RO	Over-current active : Logic 1 means that this port has an over-current condition. This bit will automatically move from one to zero when the over-current condition is removed.
5	OCC	RWC	Over-current change : Logic 1 means that the value of OCA has changed. Logic 0 means no change. SW must write a logic 1 to clear the bit
6	FPR	RW	Force Port Resume : Logic 1 means resume (K-state) detected or driven on the port. Logic 0 means no resume detected or driven on the port. The resume signaling is driven on the port as long as this bit remains a one. For legacy (L2) transitions, software must appropriately time the resume and set this bit to a zero when the appropriate amount of time has elapsed. Software does not need to time resume signaling for L1 transactions as host controller hardware will automatically enforce the necessary timing and clear this bit when the port has fully resumed. Software can influence the amount of time hardware will drive resume signaling during L1 exit via the HIRD field within the USBCMD register. This field is zero if Port Power is zero.
7	SUSP	RW	Suspend : Logic 1 means port is in the suspend state. Logic 0 means the port is not suspended. SW writes a logic 1 to this bit to put an enabled port in the L1 or L2 suspend state. Which suspend state the host controller attempts depends on the value of the Suspend Using L1 field. When in the suspend state, downstream propagation of data is blocked on this port, except for port reset. If this bit is set to a one when a transaction is in progress then the blocking will not occur until the end of the current transaction. If this bit is written to one to initiate a L1 suspend entry, the PCD interrupt status will be set upon completion of the LPM token transmission. A write of zero is ignored by the hardware. The hardware will unconditionally set this bit to zero when : - Software sets the Force Port Resume bit to zero - Software sets the Port Reset bit to a one, This field is zero if Port Power is zero or Current Connect Status is zero.
8	PR	RW	Port Reset : Logic 1 means the port is in the reset state. Logic 0 means the port is not in reset. SW writes a logic 1 to indicate the start of the reset. SW writes a logic 0 to end the reset sequence. If the reset sequence on the USB bus is finished HW will clear the bit. SW should only check the PSPD field to know the speed of the attached device when the Port Reset bit is zero. This field is zero if Port Power is zero.

Integration Guide

9	SUS_L1	RW	Suspend using L1 0b = Suspend using L2 1b = Suspend using L1 When this bit is set to a one and a non-zero value is specified in the Device Address field, the host controller will generate an LPM Token to enter the L1 state whenever software writes a one to the Suspend bit, as well as L1 exit timing during any device- or host-initiated resume. When set to zero the host controller will employ the legacy (L2) suspend mechanism. Software should only set this bit when the device attached immediately downstream of this root port supports L1 transitions.
11:10	LS	RO	Line Status : This field reflects the current logical levels of the DP (bit 11) and DM (bit 10) signal lines
12	PP	RW	Port Power : The function of this bit depends on the value of the Port Power Control (PPC) bit in the HCSPARAMS register. If PPC = 0b, this bit (PP) is read-only and will always be set to 1b. If PPC = 1b, this bit (PP) is RW, If the bit is set to zero, the port is not powered. If the bit is set to one the port is powered.
13	PFSC	RW	Port Force Full Speed Connect. Writing this bit to a 1b will force the port to only connect at full-speed. It disables the chirp sequences that allows the port to identify itself as high-speed
15:14	PIC	RW	Port Indicator Control : Writing to this field has no effect if the P_INDICATOR bit in the HCSPARAMS register is logic 0. If P_INDICATOR is set to one, these bits will indicate the value of the port indicators : 00b : Port Indicators are off 01b : Amber 10b : Green 11b : Undefined This field is zero if Port Power is zero.
19:16	PTC	RW	Port Test Control : A nonzero value indicates that the port is operating in the test mode as indicated by the value. 0000b : Test mode not enabled 0001b : Test J_STATE 0010b : Test K_STATE 0011b : TEST SE0_NAK 0100b : Test_Packet 0101b : Test Force_Enable 0110b – 1111b : Reserved The reserved values should not be written by SW.

Integration Guide

21:20	PSPD	RO	Port Speed : 00b : Low-speed 01b : Full-speed 10b : High-speed 11b : Reserved
22	WOO	RW	Wake on overcurrent enable : Writing this bit to a one enables the port to be sensitive to overcurrent conditions as wake-up events This field is zero if Port Power is zero
24:23	SUS_STAT	RO	These two bits are used by software to determine whether the most recent L1 suspend request was successful : 00b : Success – state transition was successful (ACK) 01b : Not Yet – Device was unable to enter the L1 state at this time (NYET) 10b : Not supported – Device does not support the L1 state (STALL) 11b : Timeout/Error – Device failed to respond or an error occurred. This field is updated by hardware immediately following the completion of an L1 transition request (via an LPM token). To avoid any race conditions with hardware, software should only consume the contents of this field when Suspend = 0b (port no longer in L1).
31:25	DEV_ADD	RW	Device Address for LPM tokens. 7-bit USB device address that is used when sending an LPM token to the device attached to and immediately downstream of the associated root port. A value of zero indicates no device is present or support for LPM feature is not present on this device.

4.1.2.13 ATL PTD Done Map register (Address Offset = 0x30)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	ATL_DONE	RWC	This register represents a direct map of the done status of the 32 ATL PTDs. The bit corresponding to a certain PTD will be set to logic 1 as soon as that PTD execution is completed. Writing a one to a bit in the Done Map register will clear the bit.

4.1.2.14 ATL PTD Skip Map register (Address Offset = 0x34)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	ATL_SKIP	RW	When a bit in the PTD Skip Map is set to logic 1, the corresponding PTD will be skipped, independent of the V bit setting. The information in that PTD is not

			processed. Hardware will go automatically to the next PTD.
--	--	--	--

4.1.2.15 ISO PTD Done Map register (Address Offset = 0x38)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	ISO_DONE	RWC	This register represents a direct map of the done status of the 32 ISO PTDs. The bit corresponding to a certain PTD will be set to logic 1 as soon as that PTD execution is completed. Writing a one to a bit in the Done Map register will clear the bit.

4.1.2.16 ISO PTD Skip Map register (Address Offset = 0x3C)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	ISO_SKIP	RW	When a bit in the PTD Skip Map is set to logic 1, the corresponding PTD will be skipped, independent of the V bit setting. The information in that PTD is not processed. Hardware will go automatically to the next PTD.

4.1.2.17 INT PTD Done Map register (Address Offset = 0x40)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	INT_DONE	RWC	This register represents a direct map of the done status of the 32 INT PTDs. The bit corresponding to a certain PTD will be set to logic 1 as soon as that PTD execution is completed. Writing a one to a bit in the Done Map register will clear the bit.

4.1.2.18 INT PTD Skip Map register (Address Offset = 0x44)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	INT_SKIP	RW	When a bit in the PTD Skip Map is set to logic 1, the corresponding PTD will be skipped, independent of the V bit setting. The information in that PTD is not processed. Hardware will go automatically to the next PTD.

4.1.2.19 Last PTD in use register (Address Offset = 0x48)

Reset value : 0x00000000

Bit	Symbol	Access	Description
4:0	ATL_LAST	RW	This indicates the last PTD in the ATL list. If hardware has reached this PTD and the J bit is not

Integration Guide

			set, it will go to PTD0 as the next PTD to be processed.
7:5	R	RO	Reserved
12:8	ISO_LAST	RW	This indicates the last PTD in the ISO list. If hardware has reached this PTD, it will continue with processing the INT list
15:13	R	RO	Reserved
20:16	INT_LAST	RW	This indicates the last PTD in the INT list. If hardware has reached this PTD, it will continue with processing the ATL list.
31:21	R	RO	Reserved

4.1.2.20 UTMI+/ULPI debug (Address Offset = 0x4C)

Reset value : 0x00000000

This register allows to read and write registers in the attached USB PHY. There are 2 possibilities

- USB PHY with UTMI+ interface
- USB PHY with ULPI interface

Bit	Symbol	Access	Description
7:0	PHY ADDR	RW	UTMI+ mode : Bits 3:0 are used to control VControl signal on Vendor Interface of UTMI+ ULPI mode : Bits 7:0 are used as the address when doing a register access over the ULPI interface
15:8	PHY WDATA	RW	UTMI+ mode : Reserved. This field is not used ULPI mode Bits 7:0 are used for the write data when writing a value to a ULPI PHY register
23:16	PHY RDATA	RW	UTMI+ mode : Bits 7:0 contains the value returned by the VStatus signal on Vendor Interface of UTMI+ ULPI mode Bits 7:0 are used for the read data when reading a value to a ULPI PHY register
24	PHY RW	RW	UTMI+ mode : Reserved. This field is not used ULPI mode This bit indicates if a read or write operation must be done. 0b : Read operation 1b : Write operation
25	PHY ACCESS	RW	Software writes this bit to one to start a read or write operation. HW clears the bit when the operation is finished. SW is not allowed to clear the bit.
30:26	R	RO	Reserved

Integration Guide

31	PHY MODE	RO / RW	This bit indicates if the interface between the controller is UTMI+ or ULPI 0b : UTMI+ 1b : ULPI If the hardware supports both modes, this bit is RW by SW. Otherwise it is RO. This bit will control which PHY interface is active.
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4.1.2.21 OTG Port Mode (Address Offset = 0x50)

Reset value : 0x00050000

Bit	Symbol	Access	Description
0	ID0	RO	Port 0 ID pin value. This bit indicates the value on the ID line of port 0. This field only contains a valid value some time after the ID0_EN bit is set to one. If the value of this bit changes, the PCD interrupt status bit will be set to 1b.
7:1	R	RO	Reserved
8	ID0_EN	RW	Port 0 ID pin pull-up enable. If this bit is set, the pull-up on the ID line of port 0 will be enabled. This bit must be set to read a good value in the ID0 field
15:9	R	RO	Reserved
16	PORT_MODE	RW	If this bit is set to one, the port will behave as a USB device. If this bit is set to zero, the port will be controlled by the USB host block.
17	R	RO	Reserved
18	SW_CTRL_PDCOM	RW	This bit indicates if the PHY power-down input is controlled by software or by hardware. 0b : hardware state machine controls PHY power-down 1b : software controls PHY power-down by writing to SW_PDCOM bit
19	SW_PDCOM	RW	This bit is only used when SW_CTRL_PDCOM is set to 1b. When SW_CTRL_PDCOM is set to 1b, software can directly control the PHY power-down bit by writing to this bit. 0b : PHY operational 1b : PHY in power-down mode
31:20	R	RO	Reserved

4.1.3 Proprietary Transfer Descriptor (PTD)

The standard EHCI data structures as described in “[Enhanced Host Controller Interface Specification for Universal Serial Bus Rev. 1.0](#)” are optimized for the bus controller operation that is managed by the hardware state machine.

The PTD structures of the ip_3515_hs are translations of the EHCI data structures that are optimized for the ip_3515_hs. It, however, still follows the basic EHCI architecture. This optimized form of EHCI data structures is necessary because the ip_3515_hs is a peripheral host controller and has no bus controller capability.

EHCI manages schedules in two lists: periodic and asynchronous. The data structures are designed to provide the maximum flexibility required by USB, minimize memory traffic, and reduce hardware and software complexity. The ip_3515_hs controller executes transactions for devices by using a simple shared-memory schedule. This schedule consists of data structures organized into three lists:

qISO — Isochronous transfer

qINTL — Interrupt transfer

qATL — Asynchronous transfer; for the control and bulk transfers

The system software maintains two lists for the host controller: periodic and asynchronous.

The ip_3515_hs has a maximum of 32 ISO, 32 INTL and 32 ATL PTDs. These PTDs are used as channels to transfer data from the shared memory to the USB bus. These channels are allocated and de-allocated on receiving the transfer from the core USB driver.

Multiple transfers are scheduled to the shared memory for various endpoints by traversing the next link pointer provided by endpoint data structures, until it reaches the end of the endpoint list. There are three endpoint lists: one for ISO endpoints, and the other for INTL and ATL endpoints. If the schedule is enabled, the host controller executes the ISO schedule, followed by the INTL schedule, and then the ATL schedule.

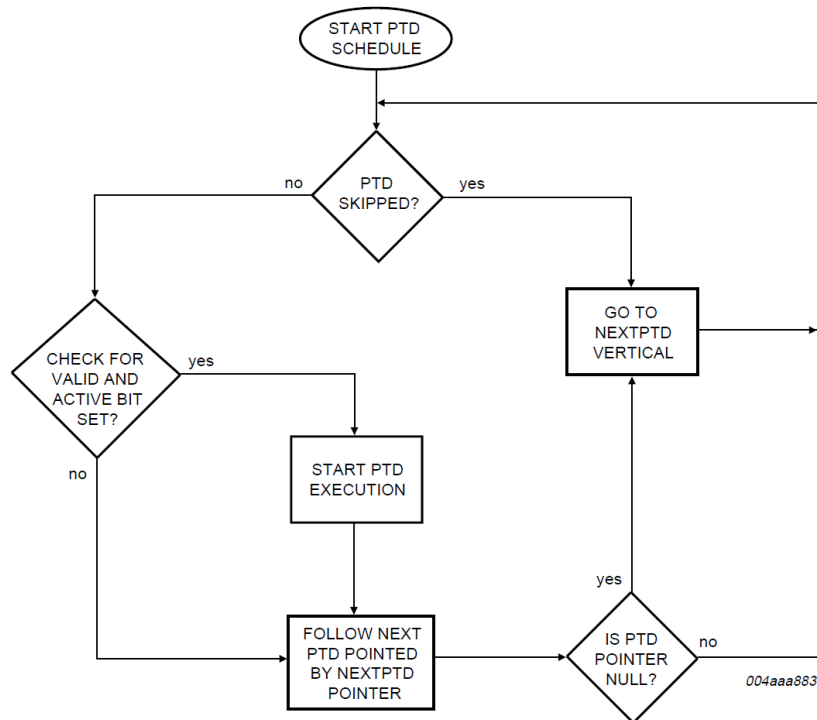
These lists are traversed and scheduled by the software according to the EHCI traversal rule. The host controller executes the scheduled ISO, INTL and ATL PTDs. The completion of a transfer is indicated to the software by the interrupt that can be grouped under various PTDs by using the AND or OR registers that are available for each schedule type: ISO, INTL and ATL. These registers are simple logic registers to decide the completion status of group and individual PTDs. When the logical conditions of the Done bit is true in the shared memory, it means that PTD has completed.

There are four types of interrupts in the ip_3515_hs : ISO, INTL, ATL and SOF. The NextPTD pointer is a feature that allows the ip_3515_hs to jump unused and skip PTDs.

This will improve the PTD transversal latency time. The NextPTD pointer is not meant for same or single endpoint. The NextPTD works only in forward direction.

The NextPTD traversal rules defined by the ip_3515_hs hardware are:

1. Start the PTD memory vertical traversal, considering the skip and LastPTD information, as follows.
2. If the current PTD is active and not done, perform the transaction.
3. Follow the NextPTD pointer as specified in bits 4 to 0 of DW4.
4. If combined with LastPTD, the LastPTD setting must be at a higher address than the NextPTD specified.
5. If combined with skip, the skip must not be set (logically) on the same position corresponding to NextPTD, pointed by the NextPTD pointer.
6. If PTD is set for skip, it will be neglected and the next vertical PTD will be considered.
7. If the skipped PTD already has a setting including a NextPTD pointer that will not be taken into consideration, the behavior will be just as described in the preceding step.



4.1.3.2 PTD on asynchronous list (qATL)

The picture below shows the bit allocation of the bulk IN and OUT, asynchronous Transfer Descriptor (ATL PTD). This data structure is used for both regular HS/FS transactions and split transactions.

PTD on Asynchronous List (normal and split transaction)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offset
R		Mult [1:0]		R	MaxPacketLength[10:0]											uFrame[7:0]							J	R	NextPTDPointer[4:0]				V	0x00		
HubAddress[6:0]					PortNumber[6:0]						SE[1:0]		RL[3:0]			S	DeviceAddress[6:0]						EP[3:0]			0x04						
DataStartAddress[15:0]															I	NrBytesToTransfer[14:0]											0x08					
A	H	B	X	SC	P	DT	Cerr[1:0]			NakCnt[3:0]			EPTtype [1:0]		Token [1:0]	NrBytesTransferred[14:0]											0x0C					

4.1.3.3 PTD on periodic list for regular transactions

The picture below shows the bit allocation of the periodic IN and OUT, periodic Transfer Descriptor. This data structure is used for regular HS/FS transactions.

PTD on Periodic List (normal)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offset
R		Mult [1:0]		R	MaxPacketLength[10:0]											uFrame[7:0]							J	R	NextPTDPointer[4:0]				V	0x00		
R														SE[1:0]		RL[3:0]			S	DeviceAddress[6:0]					EP[3:0]			0x04				
DataStartAddress[15:0]															I	NrBytesToTransfer[14:0]											0x08					
A	H	B	X	R	P	DT	Cerr[1:0]			NakCnt[3:0]			EPTtype [1:0]		Token [1:0]	NrBytesTransferred[14:0]											0x0C					
Status7[2:0]				Status6[2:0]			Status5[2:0]			Status4[2:0]			Status3[2:0]			Status2[2:0]		Status1[2:0]		Status0[2:0]		uSA[7:0]					0x10					
ISO_IN_2[7:0]								ISO_IN_1[11:0]										ISO_IN_0[11:0]							0x14							
ISO_IN_5[3:0]					ISO_IN_4[11:0]											ISO_IN_3[11:0]							ISO_IN_2[11:8]				0x18					
ISO_IN_7[11:0]										ISO_IN_6[11:0]						ISO_IN_5[11:0]						ISO_IN_4[11:0]				0x1C						

4.1.3.4 PTD on periodic list for split transactions

The picture below shows the bit allocation of the periodic IN and OUT, periodic Transfer Descriptor. This data structure is used for split transactions on the periodic list.

PTD on Periodic List (Split Transaction)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Offset
R		Mult [1:0]		R																				J	R						V	0x00
																																0x04
																																0x08
																																0x0C
																																0x10
																																0x14
																																0x18
																																0x1C

4.1.4 PTD bit definition

Symbol	Access	Description
V	SW — sets HW — resets	Valid: 0b : This bit is deactivated when the entire PTD is executed (active bit is cleared), or when a error is encountered (Halt bit is set). 1b : Software updates to one when there is payload to be sent or received. The current PTD is active.
NrBytesToTransfer[14:0]	SW — writes	Number of Bytes to Transfer: This field indicates the number of bytes that can be transferred by this data structure. It is used to indicate the depth of the DATA field (32 kB – 1 B).
MaxPacketLength[10:0] TT_MPS_Len[10:0]	SW — writes	Maximum Packet Length: This field indicates the maximum number of bytes that can be sent to or received from an endpoint in a single data packet. The maximum packet size for a bulk transfer is 512 bytes. The maximum packet size for the isochronous transfer is also variable at any whole number.
Mult[1:0]	SW — writes	Multiplier: This field is a multiplier used by the host controller as the number of successive packets the host controller may submit to the endpoint in the current execution. Set this field to 01b. You can also set it to 11b and 10b depending on your application. 00b is undefined.
I	SW — writes	Interrupt on Complete : If this bit is set and the PTD is completed, the interrupt status bit of the corresponding list is set to one.
EP[3:0]	SW — writes	Endpoint: This is the USB address of the endpoint within the function
DeviceAddress[6:0]	SW — writes	Device Address: This is the USB address of the function containing the endpoint that is referred to by this buffer
S	SW — writes	This bit indicates whether a split transaction has to be executed 0 – No split transaction – speed is same as port speed

Integration Guide

		1 – Split transaction
Token[1:0]	SW — writes	Token : Identifies the token Packet Identifier (PID) for this transaction. 00 – OUT 01 – IN 10 – SETUP 11 – Undefined
EPTType[1:0]	SW — writes	Transaction type : 00 – Control 01 – Isochronous 10 – Bulk 11 – Interrupt
SE[1:0]	SW — writes	This specifies the speed for a Control or Interrupt transaction to a device that is not high-speed : 00 – Full-speed 10 – Low-speed 01 or 11 – undefined behavior For isochronous and bulk transactions this field must be set to 00. If a full-speed hub is connected to the port, the SE[1] bit indicates if a low-speed packet must be sent on a full-speed bus.
PortNumber[6:0]	SW — writes	Port Number: This indicates the port number of the hub
HubAddress[6:0]	SW — writes	Hub Address: This indicates the hub address
NextPTDPointer[4:0]	SW — writes	Next PTD Counter: Next PTD branching assigned by the PTDpointer.
J	SW — writes	Jump: 0 — To increment the PTD pointer. 1 — To enable the next PTD branching
DataStartAddress[15:0]	SW — writes	Data Start Address: “DataStartAddress + DataPayload_BaseAddress” is the start address that points to the start of the data buffer that will be sent or received on or from the USB bus. HW does not update this field when the transfer is completed.
uFrame[7:0]	SW — writes	This field is only applicable for interrupt and isochronous endpoints. Interrupt endpoint Bits 2 to 0 of this field together with the uSA field represent the polling rate. When the polling rate is less than or equal to 1 ms. Bits 2 to 0 are set to zero. If the polling rate is greater than 1 ms. Bits 2 to 0 define the polling rate and bits 7 to 3 define the frame number when the packet must be transmitted. In case of high-speed transactions the uSA field indicates during which uFrame the packet must be sent. See table below for more information. Isochronous endpoint Bits 2 to 0 — Don't care Bits 7 to 3 — Frame number at which this PTD will be sent.

Integration Guide

NrBytesTransferred[14:0]	HW — writes	Number of Bytes Transferred: This field indicates the number of bytes sent or received for this transaction. If Mult[1:0] is greater than one, it is possible to store intermediate results in this field.
RL[3:0]	SW — writes	Reload: If RL is set to 0h, hardware ignores the NakCnt value. RL and NakCnt are set to the same value before a transaction.
NakCnt[3:0]	HW — writes SW — writes	NAK Counter: This field corresponds to the NakCnt field in TD. Software writes for the initial PTD launch. The V bit is reset if NakCnt decrements to zero and RL is a nonzero value. It reloads from RL if transaction is ACK-ed.
Cerr[1:0]	HW — writes SW — writes	Error Counter: This field corresponds to the Cerr[1:0] field in TD. The default value of this field is zero for isochronous transactions. 00 — The transaction will not retry. 11 — The transaction will retry three times. Hardware will decrement these values.
DT	HW — updates SW — writes	Data Toggle: This bit is filled by software to start a PTD. If NrBytesToTransfer[14:0] is not complete, software needs to read this value and use this value to program the next PTD that will send data to the same endpoint.
P	SW — writes HW — updates	Ping: For high-speed transactions, this bit corresponds to the Ping state bit in the Status field of a TD. 0 — Ping is not set. 1 — Ping is set. For the first time, software sets the Ping bit to 0. For the successive asynchronous TD, software sets the bit in asynchronous TD based on the state of the bit for the previous asynchronous TD of the same transfer, that is: • The current asynchronous TD is completed with the Ping bit set. • The next asynchronous TD will have its Ping bit set by the software.
SC	SW — writes 0 HW — updates	Start/Complete: 0 — Start split 1 — Complete split
X	HW — writes	Error: This bit corresponds to the Transaction Error bit in the Status field of iTD, siTD or TD. 0 — No PID error. 1 — If there are PID errors, this bit is set active. The A and V bits are also set to inactive. This transaction is retried three times.
B	HW — writes	Babble: This bit corresponds to the Babble Detected bit in the Status field of iTD, siTD or TD. 1 — When babbling is detected, A and V are set to 0
H	HW — writes	Halt: Set to a 1 by the Host Controller during status updates to indicate that a serious error has occurred at the device/endpoint addressed by this data structure.

Integration Guide

		This can be caused by babble, the error counter counting down to zero, or reception of the STALL handshake from the device during a transaction. Any time that a transaction results in the Halted bit being set to a one, the Active bit is also set to 0.
A	SW — sets	Active: This bit is the same as the Valid bit.
uSA[7:0]	SW — writes (0 → 1) HW — writes (1 → 0) After processing	This field is only used for periodic split transactions or if the port is enabled in HS mode. μSOF Active: When the frame number of bits uFrame[7:3] match the frame number of the USB bus, these bits are checked for 1 before they are sent for μSOF. For example: If μSA[7:0] = 1111 1111b: send ISO every μSOF of the entire millisecond. If μSA[7:0] = 0101 0101b: send ISO only on μSOF0, μSOF2, μSOF4 and μSOF6.
Status0[2:0]	HW — writes	Isochronous IN or OUT status at μSOF0 Bit 2 : Underrun Bit 1 : Babble Bit 0 : XactErr
Status1[2:0]	HW — writes	Isochronous IN or OUT status at μSOF1
Status2[2:0]	HW — writes	Isochronous IN or OUT status at μSOF2
Status3[2:0]	HW — writes	Isochronous IN or OUT status at μSOF3
Status4[2:0]	HW — writes	Isochronous IN or OUT status at μSOF4
Status5[2:0]	HW — writes	Isochronous IN or OUT status at μSOF5
Status6[2:0]	HW — writes	Isochronous IN or OUT status at μSOF6
Status7[2:0]	HW — writes	Isochronous IN or OUT status at μSOF7
ISO_IN0[11:0]	HW — writes	Bytes received during μSOF0, if μSA[0] is set to 1 and frame number is correct
ISO_IN1[11:0]	HW — writes	Bytes received during μSOF1, if μSA[1] is set to 1 and frame number is correct
ISO_IN2[11:0]	HW — writes	Bytes received during μSOF2, if μSA[2] is set to 1 and frame number is correct
ISO_IN3[11:0]	HW — writes	Bytes received during μSOF3, if μSA[3] is set to 1 and frame number is correct
ISO_IN4[11:0]	HW — writes	Bytes received during μSOF4, if μSA[4] is set to 1 and frame number is correct
ISO_IN5[11:0]	HW — writes	Bytes received during μSOF5, if μSA[5] is set to 1 and frame number is correct
ISO_IN6[11:0]	HW — writes	Bytes received during μSOF6, if μSA[6] is set to 1 and frame number is correct
ISO_IN7[11:0]	HW — writes	Bytes received during μSOF7, if μSA[7] is set to 1 and frame number is correct
uSCS[7:0]	SW — writes (0 → 1)	All bits can be set to one for every transfer. It specifies which μSOF the complete split needs to be sent.

Integration Guide

	HW — writes (1 → 0) After processing	Start split and complete split active bits, $\mu SA = 0000\ 0001b$, $\mu SCS = 0000\ 0100b$, will cause SS to execute in $\mu Frame0$ and CS in $\mu Frame2$.
SP_ISO_IN0[7:0]	HW — writes	Bytes received during $\mu SOF0$, if $\mu SA[0]$ is set to 1 and frame number is correct.
SP_ISO_IN1[7:0]	HW — writes	Bytes received during $\mu SOF1$, if $\mu SA[1]$ is set to 1 and frame number is correct.
SP_ISO_IN2[7:0]	HW — writes	Bytes received during $\mu SOF2$, if $\mu SA[2]$ is set to 1 and frame number is correct.
SP_ISO_IN3[7:0]	HW — writes	Bytes received during $\mu SOF3$, if $\mu SA[3]$ is set to 1 and frame number is correct.
SP_ISO_IN4[7:0]	HW — writes	Bytes received during $\mu SOF4$, if $\mu SA[4]$ is set to 1 and frame number is correct.
SP_ISO_IN5[7:0]	HW — writes	Bytes received during $\mu SOF5$, if $\mu SA[5]$ is set to 1 and frame number is correct.
SP_ISO_IN6[7:0]	HW — writes	Bytes received during $\mu SOF6$, if $\mu SA[6]$ is set to 1 and frame number is correct.
SP_ISO_IN7[7:0]	HW — writes	Bytes received during $\mu SOF7$, if $\mu SA[7]$ is set to 1 and frame number is correct.
S_Bytes	HW — writes	This field is used by HW to store an intermediate value of number of bytes received while handling a complete-split for interrupt IN transfers.

Polling rate for Periodic transactions :

The table below indicates how the hardware knows when to send a certain interrupt packet based on the polling rate. $uFrame[2:0]$ defines the polling rate.

If set to zero and the transaction is high-speed, the uSA determines during which $uFrames$ a packet can be sent. If set to zero and the transaction is a normal full-speed or low-speed, the packet will be sent during every frame. If it is set to zero and the transaction is a full-speed or low-speed split transaction, the uSA and uCS fields will determine when to send a split transaction.

If $uFrame[2:0]$ is different from zero and the transaction is high-speed, the bits in $uFrame[7:3]$ and the bits in uSA are used to know during which $uFrames$ a packet must be sent.

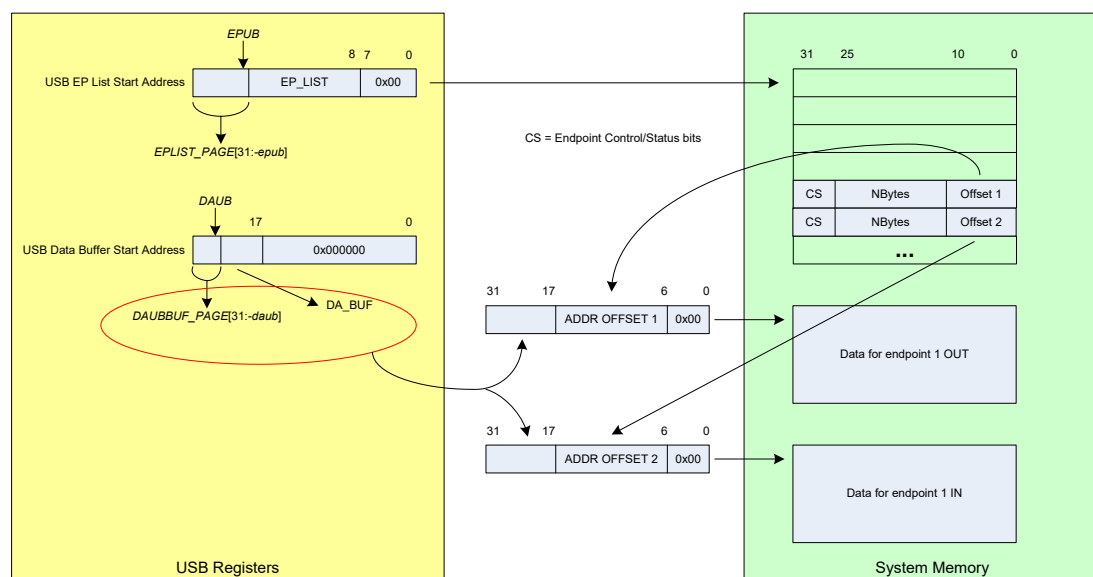
If $uFrame[2:0]$ is different from zero and the transaction is full-speed or low-speed, the bits in $uFrame[7:3]$ are used to know during which $uFrames$ a packet must be sent. If $uFrame[2:0]$ is different from zero and the transaction is a full-speed or low-speed split transaction, the bits in $uFrame[7:3]$ and the uSA and uCS fields will determine when to send a split transaction.

b	Rate	$uFrame[2:0]$	$uFrame[7:3]$	$uSA[7:0]$
1	1 μSOF	000b	Don't care	1111 1111b
2	2 μSOF	000b	Don't care	1010 1010b or 0101 0101b
3	4 μSOF	000b	Don't care	Any 2 bits set
4	1 ms	000b	Don't care	Any 1 bit set
5	2 ms	001b	Bit 0 is compared with $FRINDEX[3]$	Any 1 bit set
6	4 ms	010b	Bits[1:0] are compared with $FRINDEX[4:3]$	Any 1 bit set
7	8 ms	011b	Bits[2:0] are compared with $FRINDEX[5:3]$	Any 1 bit set
8	16 ms	100b	Bits[3:0] are compared with $FRINDEX[6:3]$	Any 1 bit set
9	32 ms	101b	Bits[4:0] are compared with $FRINDEX[7:3]$	Any 1 bit set

4.2 USB high-speed and full-speed device (IP_3511_HS)

The picture below shows which part of the data is stored in registers and what is located in system memory. The registers are explained in section 4.2.2. The variables indicated in *italic* are values that can be defined at HW compile time. They are explained in section 5.

Note : the size of the NBytes and Addr Offset fields have changed compared to ip_3511. This is to allow for bigger endpoint buffers.



4.2.1 Register overview

This section gives an overview of the register set used to control the hardware functionalities of the USB Full Speed Device Controller IP. All register addresses listed below are offsets. The base address of this IP block is system-dependent and must be defined by the integrator.

Offset	Register name	Access	Description
0x00	USB device Command/Status register	RW(C)	This register contains all the fields to control the behavior of the USB device.
0x04	USB Info register	RO	This register contains the frame number of the last received SOF, the ChipID and the error code
0x08	USB Endpoint List start address	RW	This register contains the start address of the Endpoint List that are stored in memory.
0x0C	USB Data Buffer start address	RW	This register contains the start address of the endpoint data buffers in memory.
0x10	LPM	RW	This register contains the fields for the Link Power Management support.
0x14	USB EP skip	RW	This register is used to indicate to HW that it has to deactivate the corresponding endpoint (set Active bit to zero).
0x18	USB EP Buffer in use	RW	This bit is used for double buffering. It indicates which buffer is in-use for each endpoint.

Integration Guide

0x1C	USB EP Buffer config	RW	This bit indicates if the endpoint has single buffering or double buffering.
0x20	USB Interrupt status register	RWC	This register contains the status bits of the different interrupts.
0x24	USB Interrupt enable register	RW	This register contains the enable bits of the different interrupts. If this bit is set and the corresponding interrupt status bit is set a HW interrupt is generated.
0x28	USB Set Interrupt status register	RW	If a one is written to one of the bits of this register, the corresponding interrupt status bit is set to one. When this register is read, it returns the same value as the USB Interrupt status register.
0x2C	USB Interrupt routing register	RW	Each interrupt bit has a corresponding interrupt routing bit. If the interrupt routing bit is set to zero, a HW interrupt is generated on the IRQ line if both the corresponding interrupt status and interrupt enable bits are set. If the interrupt routing bit is set to one, a HW interrupt is generated on the FIQ line if both the corresponding interrupt status and interrupt enable bits are set.
0x30	USB configuration	RO	This contains the configuration values as specified in section 5.
0x34	USB EP toggle	RO	This debug register is used to indicate the current data toggle value of the corresponding endpoint.
0x38	Reserved	RO	Reserved
0x3C	UTMI+/ULPI debug	RW	This register allows to read and write registers in the attached USB PHY.

4.2.2 Register details

4.2.2.1 USB Device Command/Status register (Address Offset = 0x00)

Reset value : 0x00000800

Bit	Symbol	Access	Description
6:0	DEV_ADDR	RW	USB device address After bus reset, the address is reset to 0x00. If the enable bit is set, the device will respond on packets for function address DEV_ADDR. When receiving a SetAddress Control Request from the USB host, firmware must program the new address before completing the status phase of the SetAddress Control Request.
7	DEV_EN	RW	USB device enable If this bit is set, the HW will start responding on packets for function address DEV_ADDR

Integration Guide

8	SETUP	RWC	SETUP token received. If a SETUP token is received and acknowledged by the device, this bit is set. As long as this bit is set all received IN and OUT tokens will be NAKed by HW. SW must clear this bit by writing a one. If this bit is zero, HW will handle the tokens to the CTRL EP0 as indicated by the CTRL EP0 IN and OUT data information programmed by SW.
9	PLL_ON	RW	Always PLL Clock on: '0' : The clock from the USB PHY will be switched off when the device is not connected or when the device is in suspend mode. At the same time, when one of these two conditions exist, the usb_needclock signal will be put low. '1' : The clock from the USB PHY will not be switched off and will be always running. The output signal usb_needclock will be always one indicating that the AHB clock cannot be switched off.
10	FORCE_VBUS	RW	If this bit is set to one, the Vbus voltage indicators from the PHY are overruled. When this bit is set, the controller will consider Vbus to be high and signal a connect when indicated by the other bits. When this bit is low, real Vbus indication are taken into account by the controller.
11	LPM_SUP	RW	LPM Supported: '0' : LPM not supported. HW returns a STALL when receiving an LPM token '1' : LPM supported.
12	IntOnNAK_AO	RW	Interrupt on NAK for interrupt and bulk OUT EP '0' : Only acknowledged packets generate an interrupt '1' : Both acknowledged and NAKed packets generate interrupts.
13	IntOnNAK_AI	RW	Interrupt on NAK for interrupt and bulk IN EP '0' : Only acknowledged packets generate an interrupt '1' : Both acknowledged and NAKed packets generate interrupts.
14	IntOnNAK_CO	RW	Interrupt on NAK for control OUT EP '0' : Only acknowledged packets generate an interrupt '1' : Both acknowledged and NAKed packets generate interrupts.
15	IntOnNAK_CI	RW	Interrupt on NAK for control IN EP '0' : Only acknowledged packets generate an interrupt '1' : Both acknowledged and NAKed packets generate interrupts.

Integration Guide

16	DCON	RW	Device status – connect The connect bit must be set by SW to indicate that the device must signal a connect. The pull-up resistor on D+ will be enabled when this bit is set and the VbusDebounce bit is one.
17	DSUS	RW	Device status – suspend The suspend bit indicates the current suspend state. It is set to '1' when the device hasn't seen any activity on its upstream port for more than 3 milliseconds. It is reset to '0' on any activity. When the device is suspended (Suspend bit = '1') and the firmware writes a '0' to it, the device will generate a remote wakeup. This will only happen when the device is connected (Connect bit = '1'). When the device is not connected or not suspended, a writing a '0' has no effect. Writing a '1' has never an effect.
18	R	RO	Reserved
19	LPM_SUS	RW	Device status – LPM Suspend This bit represents the current LPM suspend state. It is set to '1' by HW when the device has acknowledged the LPM request from the USB host and the Token Retry Time of 10us has elapsed. When the device is in the LPM suspended state (LPM suspend bit = '1') and the firmware writes a zero to this bit, the device will generate a remote wakeup. Firmware can only write a zero to this bit when the LPM_REWP bit is set to '1'. HW resets this bit when it receives a host initiated resume. HW only updates the LPM_SUS bit when the LPM_SUPP bit is equal to one.
20	LPM_REWP	RO	LPM Remote Wakeup Enabled by USB host HW sets this bit to one when the bRemoteWake bit in the LPM extended token is set to '1'. HW will reset this bit to '0' when it receives the host initiated LPM resume, when a remote wakeup is sent by the device or when a USB bus reset is received. Firmware can use this bit to check if the remote wake up feature is enabled by the host for the LPM transaction.
21	PFSC	RW	Port Force Full Speed Connect. Writing this bit to a 1b will force the port to only connect at full-speed. It disables the chirp sequences that allows the port to identify itself as high-speed
23:22	Speed	RO	This field indicates the speed at which the device operates : 00b : reserved 01b : full-speed 10b : high-speed

Integration Guide

			11b : super-speed (reserved for future use)
24	DCON_C	RWC	Device status – connect change The Connect Change bit is set when the devices pull-up resistor is disconnected because VBus disappeared. The bit is reset by firmware writing a one to it.
25	DSUS_C	RWC	Device status – suspend change The suspend change bit is set to '1' when the suspend bit toggles. The suspend bit can toggle because : - The device goes in the suspended state - The device is disconnected - The device receives resume signaling on its upstream port The bit is reset by firmware writing a one to it.
26	DRES_C	RWC	Device status – reset change This bit is set when the device received a bus reset. On a bus reset the device will automatically go to the default state (unconfigured and responding to address 0) The bit is reset by firmware writing a one to it.
27	OTG_C	RWC	On-The-Go status change This bit is set when there is a change in one of the ADP/SRP status bits. The bit is reset by firmware writing a one to it.
28	VbusDebounced	RO	This bit indicates if Vbus is detected or not. The bit rises immediately when Vbus becomes high. It drops to zero if Vbus is low for at least 3 ms. If this bit is high and the DCon bit is set, the HW will enable the pull-up resistor to signal a connect.
31:29	PHY test mode	RW	This field is written by firmware to put the PHY into a test mode as defined by the USB2.0 spec : 000b : Test mode disabled 001b : Test_J 010b : Test_K 011b : Test_SE0_NAK 100b : Test_Packet 101b : Test_Force_Enable 110b – 111b : reserved for future use

4.2.2.2 USB Info register (Address Offset = 0x04)

Reset value : 0x02000000

Bit	Symbol	Access	Description
10:0	FRAME_NR	RO	Frame number This contains the frame number of the last successfully received SOF. In case no SOF was received by the device at the beginning of a frame, the frame number returned is that of the last successfully received SOF. In case the SOF frame number contained a CRC error, the frame number returned will be the corrupted frame number as received by the device.
14:11	ERR_CODE	RW	The error code which last occurred. Upon a good packet this value is not reset. It is only set to 0x0 (No error) when an AHB reset occurs 0x0 No error 0x1 PID encoding error 0x2 PID unknown 0x3 Packet unexpected 0x4 Token CRC error 0x5 Data CRC error 0x6 Time out 0x7 Babble 0x8 Truncated EOP 0x9 Sent/Received NAK 0xA Sent Stall 0xB Overrun 0xC Sent empty packet 0xD Bitstuff error 0xE Sync error 0xF Wrong data toggle
15	R	RO	Reserved
31:16	CHIP_ID	RO	Chip identification : indicates major and minor revision of the IP - [31:24] = Major revision - [23:16] = Minor revision Major revisions used : 0x01 : USB2.0 Full-speed device 0x02 : USB2.0 Hi-speed device with UTMI interface

4.2.2.3 USB EP Command/Status List start address (Address Offset = 0x08)

Reset value : This depends on the configuration values. See chapter 6 for more information

This 32-bit register indicates the start address of the USB EP Command/Status List. Only a subset of these bits is programmable by firmware. The 8 lsb are hardcoded to zero because the list must start on a 256 byte boundary. The bits (*-epub* - 1) down to 8 can be programmed by firmware.

Note : *-epub* : this is the EP list upper bound configuration value that is defined in section 5

Integration Guide

Bit	Symbol	Access	Description
7:0	R	RO	Reserved
(-epub - 1):8	EP_LIST	RW	These are the programmable bits for firmware to indicate the start address of the USB EP Command/Status List.
31:-epub	R	RO	Reserved. This contains a hardcoded value as defined by -eplist_page[31:-epub].

4.2.2.4 USB Data buffer start address (Address Offset = 0x0C)

Reset value : This depends on the configuration values. See chapter 6 for more information

This register indicates the page of the AHB address where the endpoint data can be located

Bit	Symbol	Access	Description
16:0	R	RO	Reserved
-daub-1:17	DA_BUF	RW	These are the programmable bits for firmware to indicate the buffer pointer page where all endpoint data buffers are located.
31:-daub	R	RO	Reserved. This contains a hardcoded value as defined by -daubbuf_page[31:-daub]

Note : -daub : this is the Data Buffer upper bound configuration value that is defined in section 5

4.2.2.5 Link Power Management register (Address Offset = 0x10)

Reset value : 0x00000000

Bit	Symbol	Access	Description
3:0	HIRD_HW	RO	Host Initiated Resume Duration - HW This is the HIRD value from the last received LPM token
7:4	HIRD_SW	RW	Host Initiated Resume Duration - SW This is the time duration required by the USB device system to come out of LPM initiated suspend after receiving the host initiated LPM resume.
8	Data Pending	RW	As long as this bit is set to one and LPM supported bit is set to one, HW will return a NYET handshake on every LPM token it receives. If LPM supported bit is set to one and this bit is zero, HW will return an ACK handshake on every LPM token it receives. If SW has still data pending and LPM is supported, it must set this bit to '1'.
15:9	R	RO	Reserved

Integration Guide

16	VBus_Comp_OFF	RW	This bit is set by SW when the Session valid VBus comparator must be off. If this bit is set to one, the analog Session Valid VBus comparator will be off and VbusDebounced will be zero. If this bit is set to zero, VbusDebounced bit will function as described in the VbusDebounced register bit field.
17	CHRG_VBUS	RW	Enables charging of VBus line
18	DISCHRG_VBUS	RW	Enables discharging of VBus line
19	R	RO	Reserved
20	ADPPROBE	RO	
21	SESSION_END / ADP_SENSE	RO	
31:22	R	RO	Reserved

4.2.2.6 USB Endpoint skip (Address Offset = 0x14)

Reset value : 0x00000000

Bit	Symbol	Access	Description
29:0	SKIP	RW	Endpoint skip : Writing '1' to one of these bits, will indicate to HW that it must deactivate the buffer assigned to this endpoint and return control back to firmware. When HW has deactivated the endpoint it will clear this bit. But it will not modify the "Buffer in use" bit. An interrupt will be generated when the Active bit goes from '1' to '0'. Note : In case of double buffering, HW will only clear the Active bit of the buffer indicated by the "Buffer in use" bit.
31:30	R	RO	Reserved

4.2.2.7 USB Endpoint Buffer in use (Address Offset = 0x18)

Reset value : 0x00000000

Bit	Symbol	Access	Description
1:0	R	RO	Reserved. Fixed to zero because the control endpoint zero is fixed to single buffering for each physical endpoint.
29:2	BUF	RW	Buffer in use : This register has one bit per physical endpoint. '0' : HW is accessing buffer 0 '1' : HW is accessing buffer 1
31:30	R	RO	Reserved

4.2.2.8 USB Endpoint Buffer Config (Address Offset = 0x1C)

Reset value : 0x00000000

Bit	Symbol	Access	Description
1:0	R	RO	Reserved. Fixed to zero because the control endpoint zero is fixed to single buffering for each physical endpoint.
29:2	BUF_SB	RW	Buffer usage : This register has one bit per physical endpoint. '0' : Single buffer '1' : Double buffer If the bit is set to single buffer ('0'), it will not toggle the corresponding "USB EP Buffer in use" bit when it clears the active bit. If the bit is set to double buffer ('1'), HW will toggle the "USB EP Buffer in use" bit when it clears the Active bit for the buffer.
31:30	R	RO	Reserved

4.2.2.9 USB interrupt status register (Address Offset = 0x20)

Reset value : 0x00000000

Bit	Symbol	Access	Description
0	EP0OUT	RWC	Interrupt status register bit for the Control EP0 OUT direction. This bit will be set if NBytes transitions to zero or the skip bit is set by firmware or a SETUP packet is successfully received for the control EP0. If the IntOnNAK_CO is set, this bit will also be set when a NAK is transmitted for the Control EP0 OUT direction. Firmware can clear this bit by writing a one to it.
1	EP0IN	RWC	Interrupt status register bit for the Control EP0 IN direction. This bit will be set if NBytes transitions to zero or the skip bit is set by firmware. If the IntOnNAK_CI is set, this bit will also be set when a NAK is transmitted for the Control EP0 IN direction. Firmware can clear this bit by writing a one to it.
2	EP1OUT	RWC	Interrupt status register bit for the EP1 OUT direction. This bit will be set if the corresponding Active bit is cleared by HW. This is done in case the programmed NBytes transitions to zero, a USB OUT data packet with less bytes than the MaxPacketSize is received or the skip bit is set by firmware If the IntOnNAK_AO is set, this bit will also be set when a NAK is transmitted for the EP1 OUT direction. Firmware can clear this bit by writing a one to it.

Integration Guide

3	EP1IN	RWC	Interrupt status register bit for the EP1 IN direction. This bit will be set if the corresponding Active bit is cleared by HW. This is done in case the programmed NBytes transitions to zero or the skip bit is set by firmware. If the IntOnNAK_AI is set, this bit will also be set when a NAK is transmitted for the EP1 IN direction. Firmware can clear this bit by writing a one to it.
...			
28	EP14OUT	RWC	Interrupt status register bit for the EP14 OUT direction. This bit will be set if the corresponding Active bit is cleared by HW. This is done in case the programmed NBytes transitions to zero a USB OUT data packet with less bytes than the MaxPacketSize is received or the corresponding skip bit is set by firmware. If the IntOnNAK_AO is set, this bit will also be set when a NAK is transmitted for the EP14 OUT direction. Firmware can clear this bit by writing a one to it.
29	EP14IN	RWC	Interrupt status register bit for the EP14 IN direction. This bit will be set if the corresponding Active bit is cleared by HW. This is done in case the programmed NBytes transitions to zero or the corresponding skip bit is set by firmware. If the IntOnNAK_AI is set, this bit will also be set when a NAK is transmitted for the EP14 IN direction. Firmware can clear this bit by writing a one to it.
30	FRAME_INT	RWC	Frame interrupt. When the device is in full-speed mode, this bit is set to one every millisecond when the VbusDebounced bit and the DCON bit are set. If the device is in high-speed mode, this bit is set to one every 125 microseconds when the VBusDebounced and DCON bit are set. This bit can be used by software when handling the isochronous endpoints. Firmware can clear this bit by writing a one to it.
31	DEV_INT	RWC	Device status interrupt. This bit is set by HW when one of the device status change bits is set. Firmware can clear this bit by writing a one to it.

4.2.2.10 USB interrupt enable register (Address Offset = 0x24)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	INT_EN	RW	If this bit is set and the corresponding USB interrupt status bit is set a HW interrupt is generated on the interrupt line indicated by the corresponding USB interrupt routing bit.

4.2.2.11 USB set interrupt status register (Address Offset = 0x28)

Reset value : 0x00000000

Bit	Symbol	Access	Description
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Integration Guide

31:0	SET_INT	RW	If firmware writes a one to one of these bits, the corresponding USB interrupt status bit is set. When this register is read, the same value as the USB interrupt status register is returned.
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4.2.2.12 USB interrupt routing register (Address Offset = 0x2C)

Reset value : 0x00000000

Bit	Symbol	Access	Description
31:0	ROUTE_INT	RW	This bit can control on which hardware interrupt line the interrupt will be generated : '0' : IRQ interrupt line is selected for this interrupt bit '1' : FIQ interrupt line is selected for this interrupt bit

4.2.2.13 USB configuration (Address Offset = 0x30)

Reset value : See chapter 5 for configuration values

Bit	Symbol	Access	Description
4:0	PHYSEP	RO	Number of physical endpoints implemented in this design (excluding the default control endpoint) E.g. a value of 2 in this field indicates that the design contains the default control endpoint plus one IN endpoint (EP1 – IN) and one OUT endpoint (EP1 – OUT)
5	SB	RO	EP single buffer supported
6	DB	RO	EP double buffer supported for non-control endpoints
7	TREG	RO	Toggle register available This indicates if the Data Toggle debug register at offset 0x34 is reserved or not.
8	R	RO	Reserved
9	UTMI	RO	Indicates if the hardware supports a UTMI+ v1.0 compliant USB PHY interface
10	ULPI	RO	Indicates if the hardware supports a ULPI v1.1 compliant USB PHY interface
31:11	R	RO	Reserved

4.2.2.14 USB Endpoint toggle (Address Offset = 0x34)

Reset value : 0x00000000

This is a debug register that indicates the current value of the data toggle. It will only be there when the “*-toggle_reg*” is set to “yes” at HW compile time. If the “*-toggle_reg*” is set to “no”, this register is reserved and will return 0x00000000

Bit	Symbol	Access	Description
29:0	TOGGLE	RO	Endpoint data toggle :

Integration Guide

			This field indicates the current value of the data toggle for the corresponding endpoint.
31:30	R	RO	Reserved

4.2.2.15 UTMI+/ULPI debug (Address Offset = 0x3C)

Reset value : 0x00000000

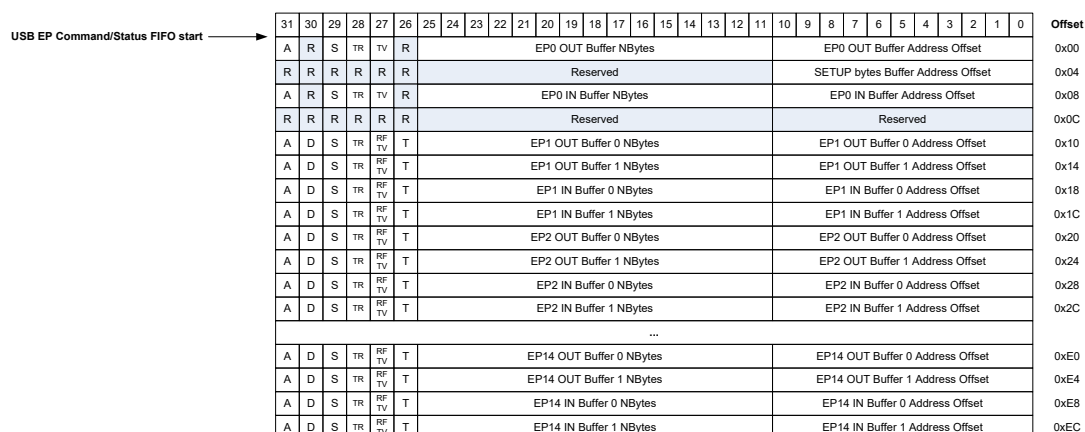
This register allows to read and write registers in the attached USB PHY. There are 2 possibilities

- USB PHY with UTMI+ interface
- USB PHY with ULPI interface

Bit	Symbol	Access	Description
7:0	PHY ADDR	RW	UTMI+ mode : Bits 3:0 are used to control VControl signal on Vendor Interface of UTMI+ ULPI mode : Bits 7:0 are used as the address when doing a register access over the ULPI interface
15:8	PHY WDATA	RW	UTMI+ mode : Reserved. This field is not used ULPI mode Bits 7:0 are used for the write data when writing a value to a ULPI PHY register
23:16	PHY RDATA	RW	UTMI+ mode : Bits 7:0 contains the value returned by the VStatus signal on Vendor Interface of UTMI+ ULPI mode Bits 7:0 are used for the read data when reading a value to a ULPI PHY register
24	PHY RW	RW	UTMI+ mode : Reserved. This field is not used ULPI mode This bit indicates if a read or write operation must be done. 0b : Read operation 1b : Write operation
25	PHY ACCESS	RW	Software writes this bit to one to start a read or write operation. HW clears the bit when the operation is finished. SW is not allowed to clear the bit.
30:26	R	RO	Reserved
31	PHY MODE	RO / RW	This bit indicates if the interface between the controller is UTMI+ or ULPI 0b : UTMI+ 1b : ULPI If the hardware supports both modes, this bit is RW by SW. Otherwise it is RO. This bit will control which PHY interface is active.

4.2.3 Endpoint command/status List

The picture below gives an overview on how the Endpoint List is organized in memory for the default configuration specified in section 5. The USB EP Command/Status List start register points to the start of the list that contains all the endpoint information in memory. The order of the endpoints is fixed as shown in the picture.



Symbol	Access	Description
A	RW	Active The buffer is enabled. HW can use the buffer to store received OUT data or to transmit data on the IN endpoint. Firmware can only set this bit to '1'. As long as this bit is set to one, firmware is not allowed to update any of the values in this 32-bit word. In case firmware wants to deactivate the buffer, it must write a one to the corresponding "skip" bit in the USB Endpoint skip register. HW can only write this bit to zero. It will do this when it receives a short packet or when the NBytes field transitions to zero or when firmware has written a one to the "skip" bit. If hardware receives a token for an endpoint that is not active, it will return the following handshake or data : - Non-isochronous endpoint : NAK handshake is sent - Isochronous OUT endpoint : received data is ignored and no handshake is sent - Isochronous IN endpoint : empty data packet is sent
D	RW	Disabled '0' : The selected endpoint is enabled. '1' : The selected endpoint is disabled. When a bus reset is received, firmware must set the disable bit of all endpoints to '1'. Firmware can only modify this bit when the active bit is zero.
S	RW	Stall '0' : The selected endpoint is not stalled '1' : The selected endpoint is stalled The Active bit has always higher priority than the Stall bit. This means that a Stall handshake is only sent when the active bit is zero. Firmware can only modify this bit when the active bit is zero.

Integration Guide

TR	RW	Toggle Reset When firmware sets this bit to one, the HW will set the toggle value equal to the value indicated in the “toggle value” (TV) bit. For the control endpoint zero, this is not needed to be used because the hardware resets the endpoint toggle to one for both directions when a setup token is received. For the other endpoints, the toggle can only be reset to zero when the endpoint is reset.
RF / TV	RW	Rate Feedback mode / Toggle value For the control endpoint zero this bit is used as the toggle value. When the toggle reset bit is set, the data toggle is updated with the value programmed in this bit. For the non-control endpoints, this bit is used together with the T-bit to identify the type of endpoint When the endpoint type (T) is set to generic endpoint, this bit selects between bulk endpoint and interrupt endpoint in rate-feedback mode. ‘0’ : Bulk endpoint with maximum packet size of 512 bytes in HS mode and 64 bytes in FS mode ‘1’ : Interrupt endpoint in ‘rate feedback mode’. This means that the data toggle is fixed to zero for all data packets. When the interrupt endpoint is in ‘rate feedback mode’, the TR bit must always be set to zero. When the endpoint type (T) is set to periodic, this bit determines if the endpoint is interrupt or isochronous. ‘0’: Isochronous endpoint (Max Packet Size is determined by the smallest value when comparing NBytes field with 1024) ‘1’: Interrupt endpoint (Max Packet Size is determined by the smallest value when comparing NBytes field with 1024)
T	RW	Endpoint Type ‘0’ : Generic endpoint. The endpoint is configured as a bulk or rate feedback interrupt endpoint. In case of an rate feedback interrupt endpoint, the Maximum Packet Size in High-Speed mode can only be maximum 512 bytes. ‘1’ : Periodic endpoint. The RF / TV bit determines if the endpoint is isochronous or interrupt.
NBytes	RW	For OUT endpoints this is the number of bytes that can be received in this buffer. For IN endpoints this is the number of bytes that must be transmitted. HW decrements this value with the packet size every time when a packet is successfully transferred. Note : If a short packet is received on an OUT endpoint, the active bit will be cleared and the NBytes value indicates the remaining buffer space that is not used. Firmware calculates the received number of bytes by subtracting the remaining NBytes from the programmed value.

Integration Guide

Address Offset	RW	Bits 16 downto 6 of the buffer start address. This address offset is updated by HW after each successful reception/transmission of a packet. HW increments the original value with the rounded up integer value when the packet size is divided by 64. E.g. if a packet of 200 bytes is successfully received, the Address Offset will be incremented by 4. If a packet of 64 bytes is successfully received, the Address Offset is incremented by 1. If a packet of less than 64 bytes is received, the Address Offset is also incremented by 1. If a packet with zero bytes is received, the Address Offset is not incremented.
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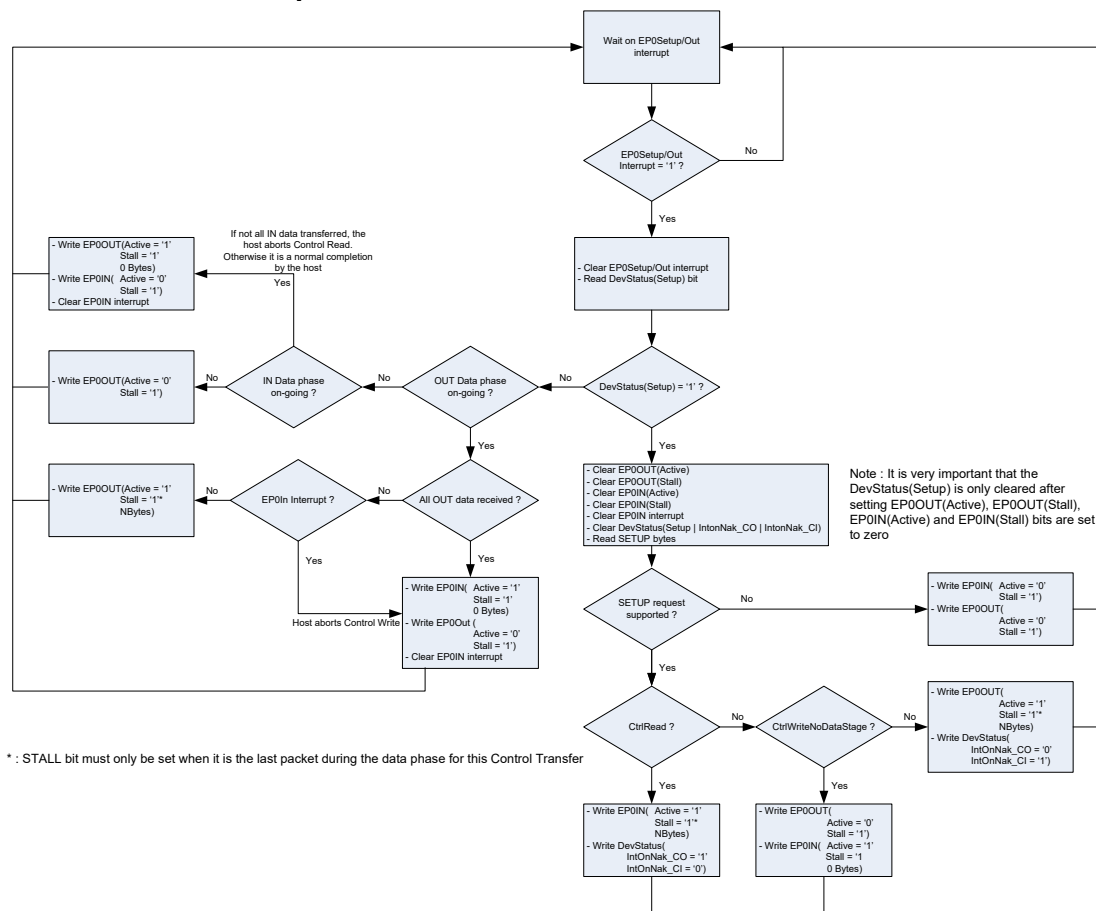
Note : When receiving a SETUP token for endpoint zero, the HW will only read the SETUP bytes Buffer Address offset to know where it has to store the received SETUP bytes. HW will ignore all other fields. In case the SETUP stage contains more than 8 bytes, it will only write the first 8 bytes to memory. A USB compliant host must never send more than 8 bytes during the SETUP stage.

4.2.4 Firmware flow charts

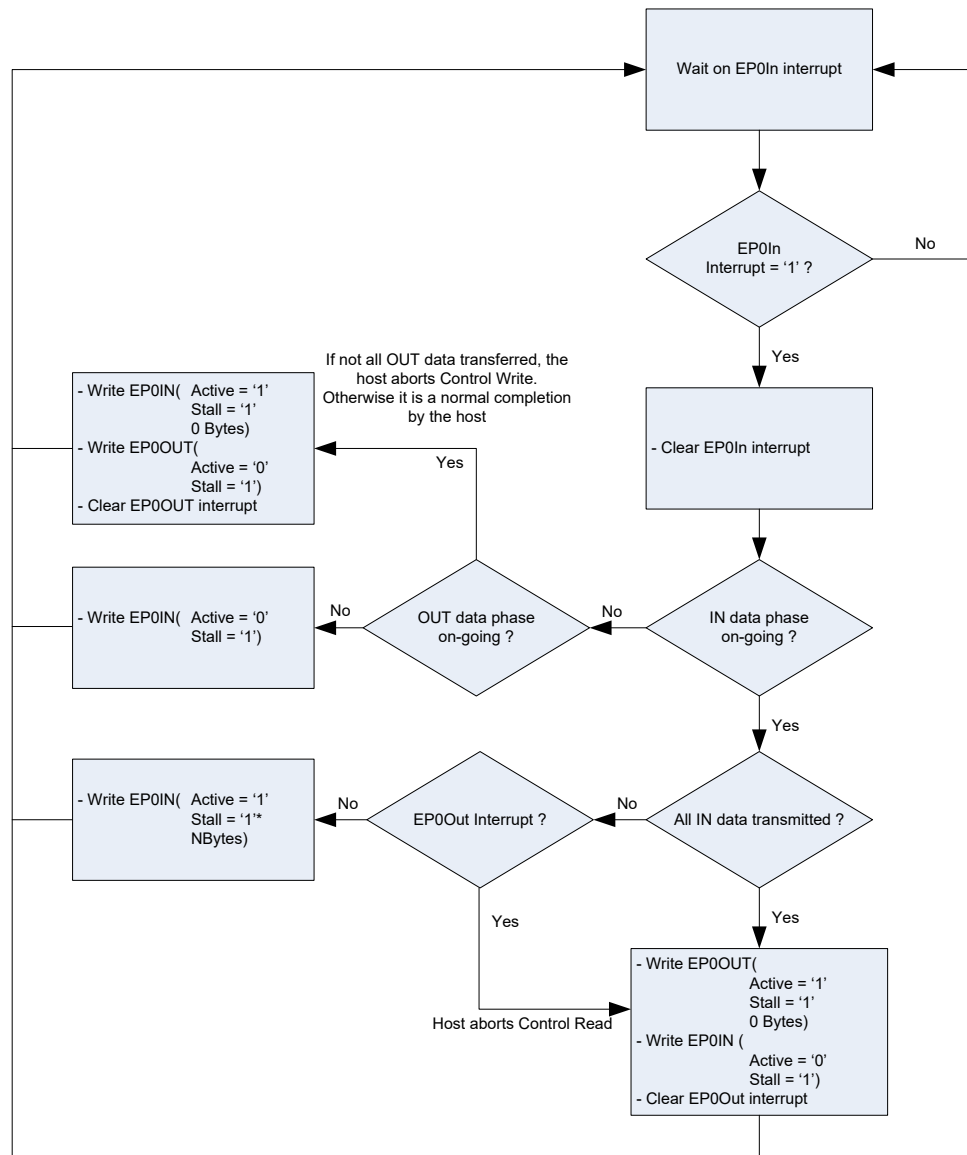
The flow charts in the sections below indicate how firmware can handle the different endpoints.

4.2.4.1 Control endpoint zero

4.2.4.1.1 Control endpoint zero – OUT direction



4.2.4.1.2 Control endpoint zero – IN direction



* : STALL bit must only be set when it is the last packet during the data phase for this Control Transfer

4.2.4.2 Generic endpoint

4.2.4.2.1 Generic endpoint – single buffering

To enable single buffering, firmware must set the corresponding “USB EP Buffer Config” bit to zero. In the “USB EP Buffer in use” register, firmware can indicate which buffer is used in this case.

When firmware wants to transfer data, it programs the different bits in the Endpoint command/status entry and sets the active bits. HW will transmit/receive multiple packets for this endpoint until the NBytes value is equal to zero. When NBytes goes to zero, HW clears the active bit and sets the corresponding interrupt status bit.

Firmware must wait until HW has cleared the Active bit to change some of the command/status bits. This prevents that HW will overwrite a new value programmed by firmware with some old values that were still cached.

If firmware wants to disable the active bit before HW has finished handling the complete buffer, it can do this by setting the corresponding endpoint skip bit in USB endpoint skip register.

4.2.4.2.2 Generic endpoint – double buffering

To enable double buffering, firmware must set the corresponding “USB EP Buffer Config” bit to one. The “USB EP Buffer in use” register indicates which buffer will be used by HW when the next token is received.

When HW clears the active bit of the current buffer in use, it will switch the buffer in use.

Firmware can also force HW to use a certain buffer by writing to the “USB EP Buffer in use” bit.

4.2.4.3 Special cases

4.2.4.3.1 Usage of the Active bit

The use of the Active bit is a bit different between OUT and IN endpoints.

When data must be received for the OUT endpoint, the firmware will set the Active bit to one and program the NBytes field to the maximum number of bytes it can receive.

When data must be transmitted for an IN endpoint, the firmware sets the Active bit to one and programs the NBytes field to the number of bytes that must be transmitted.

4.2.4.3.2 Generation of a STALL handshake

Special care must be taken in programming the endpoint to send a STALL handshake. A STALL handshake is only sent when

- The endpoint is enabled (Disabled bit = '0')
- The Active bit of the endpoint is set to zero. (No packet needs to be received/transmitted for that endpoint)

4.2.4.3.3 ClearFeature(Endpoint Halt)

When a non-control endpoint has returned a STALL handshake, the host will send a ClearFeature(Endpoint Halt) for that endpoint. When the device receives this request, the endpoint must be unstalled and the toggle bit for that endpoint must be reset back to zero. In order to do that the firmware must program the following items for the endpoint that is indicated.

In case the endpoint is used in single buffer mode :

- Write STALL bit (S) to zero
- Write toggle reset bit (TR) to one and write toggle value bit (TV) to zero

When the endpoint is used in double buffer mode :

- Write STALL bit (S) of both buffer 0 and buffer 1 to zero

- Read the Buffer in Use bit for this endpoint
- Write toggle reset bit (TR) to one and write toggle value bit (TV) to zero for the buffer indicated by the Buffer in Use bit.

4.2.4.3.4 SetConfiguration

When a SetConfiguration request is received with a configuration value different from zero, the device firmware must enable all endpoints that will be used in this configuration and reset all the toggle values. To do so, it must generate the procedure explained in section 5.4.3.1 ClearFeature(Endpoint Halt) for every endpoint that will be used in this configuration.

For all endpoints that are not used in this configuration, it must set the Disabled bit (D) to one.

5. Configuration Interface

The table below lists the configuration options for this IP. These options can be set by setting the constant values using the generic map available at the toplevel.

Option	Description	Config space	Notes
-ram_addresswith	Number of bits for the address bus of the RAM	15:9 Default : 9	This defines the size of the attached RAM. The RAM has a fixed data width of 64 bits.
-nbgenphysep	Number of physical endpoints	0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 Default : 28	This value indicates the number of physical bulk, interrupt or isochronous endpoints that are supported by the device. This value excludes the default Control endpoint 0 The first generic endpoint starts at logical endpoint number 1. ¹
-singlebuff	Single buffering supported	Yes No Default : Yes	This indicates if single buffer must be supported for the endpoints.
-doublebuff	Double buffering supported	Yes No Default : Yes	This indicates if double buffer must be supported for the endpoints.
-epub	EP List Upperbound	32:8 Default : 32	This value minus one indicates the msb of the EP command/status list start address that is still programmable by firmware. The bits at a position equal or higher than this value will have a fixed value that can be set before compile time by the <code>eplist_page</code> value.
-daub	DATA buffer page upper bit	32:-dalb Default : 32	This value minus one indicates the msb of the DATA buffer page that is still programmable by firmware. The bits at a position equal or higher than this value will have a fixed value that can be set before compile time by the <code>databuf_page</code> value.
-dalb	DATA buffer page lower bit	17:7 Default : 17	This value indicates the lsb of the DATA buffer page that is still programmable by firmware. The bits below this position can be programmed for each endpoint separately using the address offset field in the EPConfig list.
-toggle_reg	Data toggle register available	Yes No Default : Yes	This indicates if the data toggle debug register is available or not.
-eplist_page	EP List Page pointer	32 bit value	This defines the msb values for the Endpoint list address. The number of msb bits that is used is determined by the <code>-epub</code> configuration value
-databuf_page	Data Buffer Page pointer	32 bit value	This defines the msb values for the Data buffer address. The number of

¹ 1 logical endpoint consists of 1 physical OUT endpoint and 1 physical IN endpoint

Integration Guide

			msb bits that is used is determined by the –daub configuration value
--	--	--	--

Abbreviations

Abbreviation	Description
FS	Full speed
LS	Low speed
HS	High speed
UTMI+	USB2.0 Transceiver Macro cell Interface Plus
ULPI	UTMI+ Low pin interface
USB	Universal Serial Bus
SE0	Single ended 0
SE1	Single ended 1
EOP	End of Packet
SOF	Start of Frame

Integration Guide

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8. Contents

1.	Introduction	3
1.1	Purpose	3
1.2	Scope	3
1.3	References	3
2.	Architecture overview.....	4
3.	Hardware Interface.....	5
3.1	Pin Description	5
3.2	Detailed timing of interfaces.....	12
4.	Software Interface.....	13
4.1	USB high-speed, full-speed and low-speed host (IP_3515_HS).....	13
4.1.1	Register overview	14
4.1.2	Register details	15
4.1.3	Proprietary Transfer Descriptor (PTD).....	27
4.1.4	PTD bit definition.....	29
4.2	USB high-speed and full-speed device (IP_3511_HS).....	34
4.2.1	Register overview	34
4.2.2	Register details	35
4.2.3	Endpoint command/status List.....	46
4.2.4	Firmware flow charts	49
5.	Configuration Interface.....	53
	Abbreviations	54
6.	Legal information	55
6.1	Disclaimers	55
6.2	Trademarks	56
7.	Contact information	57
8.	Contents	57

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Date of release: 5 March 2026

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